Bank Branches and the Allocation of Capital across Cities*

Olivia Bordeu† **Gustavo González**‡ **Marcos Sorá**§

April 13, 2026

Click here for the latest version.

Abstract

We study how the location of bank branches and imperfect competition in credit markets affect the allocation of capital across cities, using data from Chile. We show that bank-level deposit shocks increase lending and lower interest rates, with stronger effects in more competitive local markets. We develop and estimate a quantitative spatial model with oligopolistic banks and frictional interbank lending, and find that spatial misallocation from market power reduces GDP by 0.52 pp while the impact of interbank frictions is negligible. Bank mergers create a trade-off between improved spatial financial integration and the welfare costs of reduced competition, which are stronger if the merging banks overlap in many cities.

Key words: bank branches, local credit markets, misallocation.

JEL codes: D43, G21, O16.

*We are grateful to Rodrigo Adão, Fernando Cirelli, Jonathan Dingel, Santiago Franco, Quinn Maingi, Eduardo Morales, Esteban Rossi-Hansberg, Pedro Teles, Daniel Xu, and audiences at the Banco Central de Chile, Banco de Portugal, Católica Lisbon, Duke, Nova SBE, Princeton, the University of Chicago and numerous conferences for helpful comments and suggestions. Esteban Aguilera, Sebastián Andalaft and Valeria Logan provided superb research assistance. We thank Alessandro Gavazza for sharing estimates from Albertazzi et al. (2024). An earlier version was circulated under the title "The Spatial Distribution of Bank Branches and Domestic Capital Flows." The views and conclusions presented in the paper are exclusively those of the authors and do not necessarily reflect the position of the Central Bank of Chile or of the Board members.

†University of California at Berkeley, Haas School of Business, email: oliviabordeu@berkeley.edu

‡Pontificia Universidad Católica de Chile, email: gggonzal@uc.cl

§University of Illinois at Urbana-Champaign, email: msora@illinois.edu

1 Introduction

In modern economies, savings are held largely as digital money and can be transferred across locations at virtually no cost. It is therefore natural to conjecture that savings flow toward cities with high loan demand, arbitraging away differences in the return to capital across locations. Yet, evidence from a series of studies suggests that local credit supply is not perfectly elastic and instead depends on features of local bank branches (Petersen and Rajan, 2002; Gilje et al., 2016; Nguyen, 2019; Bustos et al., 2020; Aguirregabiria et al., 2025).

We take the results from this literature as our starting point and study how the location of bank branches and banking market structure limits spatial arbitrage and, therefore, the efficient allocation of capital across cities. We make two main contributions. First, we extend results in the empirical literature by studying how banks respond to deposit shocks by lending more and reducing interest rates following a similar empirical strategy than Gilje et al. (2016) and Bustos et al. (2020). We also document heterogeneous responses across cities, with pass-through being larger in cities where banks face more competition. Second, we build a quantitative spatial model with banks that can replicate our reduced-form estimates. In both classical and recent studies of the spatial distribution of economic activity it has generally been assumed that interest rates equalize across cities (Henderson, 1974; Acemoglu and Dell, 2010; Desmet and Rossi-Hansberg, 2013; Kleinman et al., 2023). Our model bridges the gap between these two strands of the literature and allows us to study the role of bank branches in general equilibrium.

Using data from Chile, we first show that bank-level deposit inflows lead to more loan issuance and reductions in lending rates relative to other banks. We construct exogenous bank-level deposit shocks by leveraging changes in the world price of salmon interacted with banks' presence in regions specialized in fishing. We estimate that a 1% increase in deposits leads to an increase in loans of 0.11% and a reduction in interest rates of 0.08 basis points at the bank level. Our results align with Gilje et al. (2016) and Bustos et al. (2020), who study commodity shocks in the U.S. and Brazil but do not examine interest rates due to data constraints. We overcome this limitation by exploiting administrative loan-level data, which includes information on interest rates. The data allows us to control for changes in the composition of loans across time and isolate the effect of deposit shocks on interest rates.

We document heterogeneous responses to deposit shocks across cities. The positive effect on loans and the negative effect on interest rates do not decay with distance from the shocked area, suggesting that within-bank capital flows are independent from geography. By contrast, the effect of deposit shocks on loans and interest rates is stronger in cities where the exposed bank held a small market share. The intuition is

that banks with greater local market power absorb part of the cost reduction by increasing their markups, dampening the pass-through to borrowers. These results align with studies in industrial organization and finance highlighting the local nature of credit markets (Aguirregabiria et al., 2025) and motivate our focus on local market power as a determinant of differences in credit supply across cities.

In the second part of the paper, we build a quantitative spatial model with banks that allows us to quantify how the geographical distribution of bank branches affect interest rates across space and the spatial allocation of capital in Chile. We extend the framework of Kleinman et al. (2023), which includes trade, migration, and local investment decisions by introducing a set of nationally chartered banks. These banks operate branches across cities, where they offer savings and lending instruments to the local population which uses the latter to finance local investment. We incorporate an interbank market in which banks can lend to each other, subject to frictions. Local branches compete oligopolistically in each credit market, as in the trade model of Atkeson and Burstein (2008). Through this mechanism, cities with stronger competition have lower equilibrium interest rates. While banks' pricing decisions are interdependent due to local market structure and interactions in the interbank market, the steady state computation of the model remains tractable.

As in any model with imperfect competition, the elasticity of loan demand with respect to each bank's interest rate plays a central role. This elasticity, which varies across cities and banks, captures two margins of adjustment available to capitalists when a bank raises interest rates: capitalists can substitute by borrowing from other banks without changing how much they invest in physical capital, or they can reduce investment and rely more on deposits to transfer resources inter-temporally. This second elasticity —which we call the outer elasticity — plays the role of the outer-nest elasticity in the well-known model of variable markups from Atkeson and Burstein (2008). Importantly, the outer elasticity is an endogenous object in our model and depends on capitalists' demand for deposits.

We estimate the model to match the elasticity of loan issuance and interest rate reductions to deposit shocks from our empirical analysis, as well as employment, wages, and loans across cities and banks in 2015. In the model, the effect of deposit shocks on loan issuance is closely linked to the size of interbank frictions, while interest rate reductions are closely linked to firms' elasticity of substitution between banks. In our quantified version of the model, we estimate that interbank frictions raise borrowing costs by approximately 13 basis points for the average bank, i.e: a bank borrowing in the interbank market for a rate of 1% would act as if the interest rate was approximately 1.13%. We estimate an elasticity of substitution of 14.1, which, when combined with our estimated outer elasticities, yields average markups on loan interest rates of

approximately 0.09 pp.

In the model, interest rates differ across cities due to two channels. Frictions in the interbank market imply that banks with better access to deposits have a lower cost of raising funds. Differences in interest rates across banks translate into differences in interest rates across cities due to banks' heterogeneous geographic presence. Furthermore, the interest rate charged by any bank may differ across cities due to differences in local competition. Banks set lower markups in markets where they face more competition and, all else equal, interest rates are lower in cities with more banks. In the quantified version of the model, the standard deviation in interest rate across cities is 170 basis points, while the gap between the 25th and 75th percentiles is 168 basis points. While local interest rate levels are not a target of our estimation, these magnitudes align with the empirical findings of Bordeu et al. (2025) using the same data.

Equipped with the quantified version of the model, we solve for the equilibrium of this economy with no interbank frictions, which leads to an equalization in the cost of raising funds across banks. Productivity in the economy increases by 0.02%. Using subsidies to correct banks' local market power would have a substantially larger impact on GDP. Completely eliminating markups would increase aggregate GDP by 3.55%, while equalizing markups across space would increase GDP by 0.52%. In both cases, the increase in GD reflects lower dispersion in the marginal productivity of capital across cities, as well as higher overall investment, which is reflected in a lower average marginal productivity of capital.

Finally, we use the model to analyze bank mergers, a frequent challenge for policy makers — in Chile alone, there were four large mergers during 2000-2020 (Marivil et al., 2021). We compute all possible two-bank mergers and find welfare effects ranging between -1.7% and 0%. Welfare effects are larger when the merging banks have limited geographic overlap, which reduces the effect of the merger on markups, and when the two merging banks have different positions in the interbank market. In such cases, the merged entity can exploit internal capital markets, bypassing the frictions in the interbank market.

This paper contributes to the literature on economic linkages across space by characterizing financial linkages driven by the branch network. The literature has emphasized mechanisms such as trade, migration, and commuting, through which local shocks affect nearby locations via market access or labor flows (Caliendo et al., 2017; Monte et al., 2018; Allen and Arkolakis, 2025). By contrast, the mechanism we highlight operates independently of geographic proximity: through the banking network, distant cities can become linked when they share branches of the same banks (Gilje et al., 2016; Bustos et al., 2020; Maingi, 2026; D'Amico and Alekseev, 2024; Quincy and Xu, 2025). Kleinman et al. (2023) incorporates capital into a quantitative spatial

model, but does not model financial linkages between cities.[1]

Recent work has incorporated banks into quantitative spatial models to study the effects of U.S. branching deregulation and deposit reallocation (D'Amico and Alekseev, 2024; Oberfield et al., 2024; Maingi, 2026). A closely related paper is Maingi (2026), which analyzes how deposit reallocation across banks shapes lending across cities. In contrast to these studies, we focus on how the geographic distribution of branches affects steady-state outcomes, including productivity and welfare, using a model with endogenous investment and migration decisions. The financial block of our model is different in two dimensions: we incorporate oligopolistic competition in local credit markets and market-clearing in the interbank market. After laying out the model, we discuss the role of each of these channels in Section 4.3. The focus of our policy analysis, bank mergers, is also different. Corbae and D'Erasmo (2025) build a quantitative model of bank dynamics with imperfect competition but where the fully-fledged spatial component is absent.

Finally, we contribute to the literature in industrial organization and finance that studies the role of geography in lending. Petersen and Rajan (2002) document how advances in information technology allowed U.S. borrowers to access more distant lenders. Ashcraft (2005), Garmaise and Moskowitz (2006) and Nguyen (2019) show that bank branch closures led to declines in lending, while Becker (2007) and Gilje et al. (2016) show that local bank-level deposit shocks lead to loan growth by exposed banks. Taken together, these studies suggest that despite technological change, credit markets remain local to some extent. In line with this definition of the market, a series of papers have examined banks' local market power in segmented credit markets (Degryse and Ongena, 2005; Scharfstein and Sunderam, 2016; Drechsler et al., 2017; Crawford et al., 2018; Wang et al., 2020; Aguirregabiria et al., 2025; Bordeu et al., 2025). We contribute to this literature by quantifying the general equilibrium effects of local market power in a spatial model.

The rest of the paper is organized as follows. In Section 2, we overview the banking sector and its geographic footprint in Chile, and describe our data sources. In Section 3, we analyze localized deposit shocks and trace their impact on loans and interest rates across cities. Section 4 presents our quantitative model and Section 5 our estimation procedure. In Section 6, we use the model to explore the economic effects of interbank frictions and market power, while in Section 7 we simulate all possible two-bank mergers in Chile. Section 8 concludes.

[1] A vast literature, including Lucas (1990), study frictions to international investment. See Pellegrino et al. (2025) for a recent framework to analyze frictions in an international context.

2 Context and data sources

Chile has a well-developed financial system in which banks play a central role. Between 2012 and 2018, the period we analyze, credit to the private sector increased from 104% to 118% of GDP. Banks account for roughly 78% of credit, and survey data show that firms of all sizes rely heavily on banks for financing.[2] Given our focus on spatial disparities, it is noteworthy that despite the overall depth of Chile's financial system, financial development varies markedly across Chilean regions. Figure 1a illustrates this by showing outstanding bank loans in 2015, scaled by regional GDP.[3]

The Chilean banking industry is highly concentrated at the national level. Panel A in Table 1 reports loan market shares for the ten largest banks. The two largest banks each held approximately 18% of the loan market; the top four accounted for just over 60% of the market; and the ten largest banks together covered nearly the entire market. Omitting *BancoEstado*, the only state-owned retail bank, the Herfindahl–Hirschman Index (HHI) in 2015 was 1,600.

Banks in Chile are chartered nationally, headquartered in Santiago, and operate branches across the country. The first column of Panel A of Table 1 shows the number of cities served by each bank. We find no evidence of spatial correlation between a bank's activity on the extensive margin (presence in a city) or on the intensive margin (local market share) across city pairs at varying distances. Following Conley and Topa (2002), we interpret this as evidence that branches are not geographically clustered.[4] This pattern stands in contrast to the U.S., where historical restrictions on interstate expansion have led banks to concentrate regionally (Oberfield et al., 2024). The absence of geographic clustering in Chilean banking makes this an ideal setting to isolate the role of financial linkages from that of physical proximity.

Cities tend to be served by a small number of banks. The average number of banks per city was 4.4, and the median was 3. Figure 1b plots, for each region, the median number of banks per city. There is a positive correlation (0.38) between the median number of banks per city and regional financial development, discussed before. The small number of banks translates into an average HHI across cities of 3,300 and a median of 2,700.[5] Figure 1c displays the distribution of local HHI across regions, and the second column in Panel B in Table 1 shows summary statistics at the city-level.

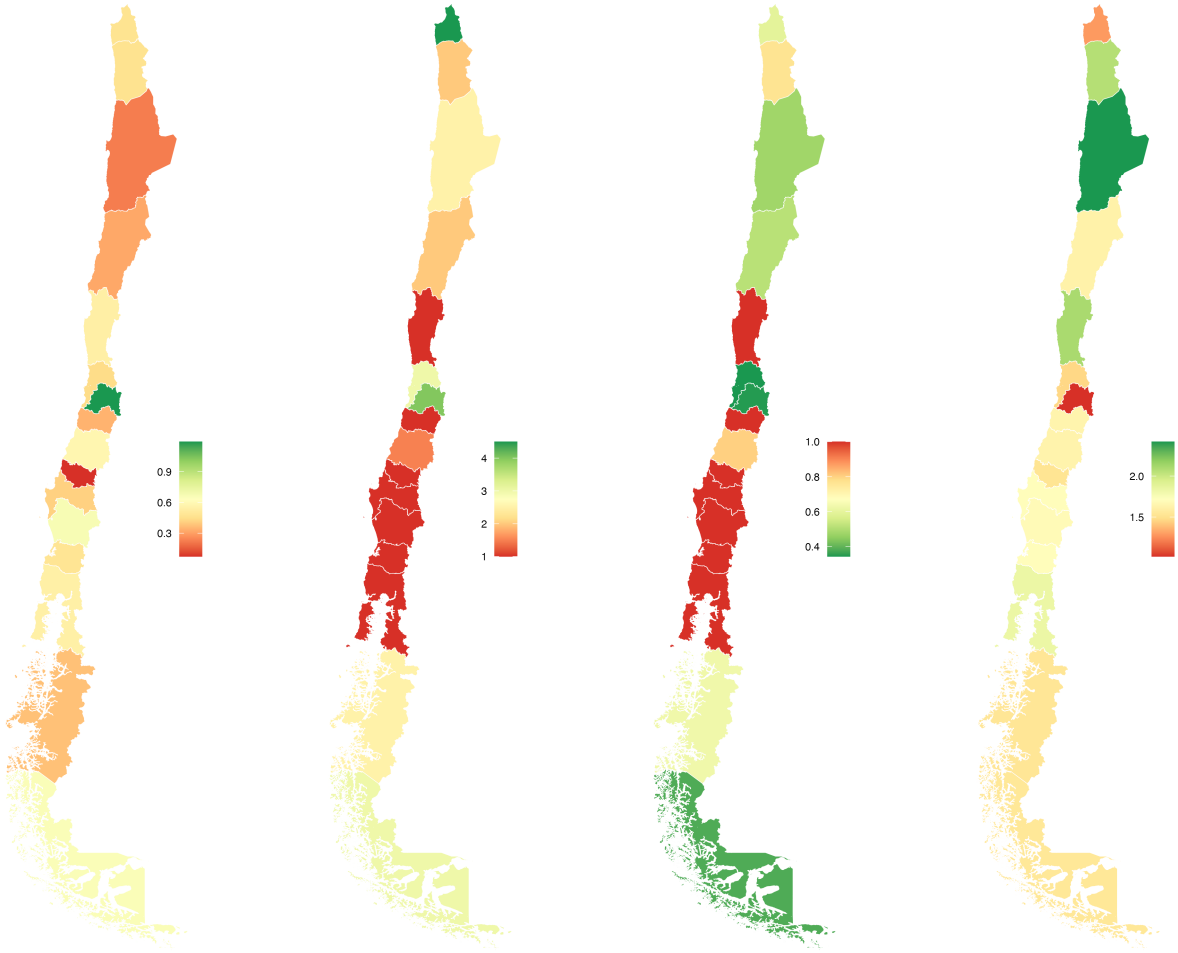
Banks shift resources across cities by using deposits collected in one location to finance loans elsewhere.

[2] See Appendix Section A.1 and Section A.2 for a discussion of the empirical references in this paragraph.

[3] Regions are the finest level of disaggregation at which GDP data are available. This measure of financial development excludes credit extended by non-bank financial institutions and instruments other than loans, so the resulting ratios are lower than those reported nationally.

[4] See Section A.4 in the Appendix for a detailed description of these results.

[5] These indices exclude very small cities with only one bank, typically *BancoEstado*.



(a) Financial Development (b) Banks per City (c) HHI in Loans (d) Loans-to-Deposits

Figure 1: Spatial Financial Development and the Network of Bank Branches

Notes: Authors' calculations using public data from the CMF on loans and deposits and Regional GDP data from the Central Bank of Chile. The first panel shows the value of loans issued during 2015 in each region over regional GDP, both in current prices. For the third and fourth panels, we use the stock of outstanding loans and deposits in December 2015.

To measure the extent of domestic capital flows, we calculate the loan-to-deposit ratio for each city using the stock of outstanding loans and deposits as of December 2015. Figure 1d shows the resulting pattern at the regional level and the third column in Panel B of Table 1 at the city level. There is substantial variation in the relative importance of deposits and loans across cities.

Although banks' liabilities also include bonds, external credit, and Central Bank borrowing through credit lines, deposits remain banks' primary funding source, while loans account for the majority of their assets (Marivil et al., 2021). As of December 2015, the system-wide loan-to-deposit ratio stood at 1.14. Panel A of Table 1 documents how this ratio varies across banks. In our subsequent analysis we abstract from alternative instruments and allow banks to lend to each other to capture dispersion in loan-to-deposit ratio across banks.

Table 1: Summary Statistics: The Spatial Distribution of Banks in 2015

A. Top banks	City Coverage	National Loan Share	Loans-to-Deposits
Santander	97	0.18	1.18
de Chile	111	0.18	1.27
BancoEstado	223	0.14	0.89
de Crédito e Inversiones	93	0.12	1.11
BBVA	61	0.06	1.21
Corpbanca	47	0.06	1.17
Scotiabank	51	0.06	1.54
Itaú	38	0.05	1.33
Security	17	0.03	1.12
BICE	16	0.02	1.03
B. Cities	Banks per City	HHI in Loans	Loans-to-Deposits
Average	4.4	3,300	0.92
Percentile 25	2	1,900	0.40
Median	3	2,700	0.66
Percentile 75	7	4,100	1.36

Notes: Authors' calculations using public data from the CMF. We use the stock of outstanding loans and deposits in December 2015 for the second and third columns of panel B.

2.1 Data sources

Our first source of data is publicly available data on total deposits and loans at the city-bank level from the Financial Market Commission (henceforth CMF), the agency responsible for supervising the stability and development of Chile's financial markets. The CMF collects detailed reports from financial institutions, including the stock of loans and deposits by type, currency, bank, and city. We construct city-bank aggregates by adding up instruments denominated in local currency, inflation-adjusted units, and foreign currency, combining both commercial and mortgage loans, as well as deposits with varying liquidity. The value of loans and deposits at the city-bank level in 2015 forms the basis of descriptive statistics in Section 2, the empirical analysis focused on quantities in Section 3, and are part of our estimation targets in Section 5. For the last two exercises we exclude loans and deposits from *BancoEstado*. The data covers 2012-2018 and is reported monthly.

To analyze the effect of deposit shocks on interest rates we complement this data with administrative loan-level data for the same period, also collected by the CMF. Given that interest rates differ across instruments and borrowers, the main advantage of this data is that we can focus on loans taken by firms, while the value of loans reported in the publicly available data includes loans for all types of purposes such as mortgages or consumer loans. Moreover, the richness of the data allows us to control for a rich set of characteristics of the borrowing firm and the loan, which is crucial for interpreting changes in average interest rates across

time. Section A.5 in the Appendix contains detail on the variables and how we restrict our sample. We use these data for the empirical analysis focused on interest rates in Section 3, also omitting loans issued by *BancoEstado*.

In Section 5, we estimate a subset of parameters from our quantitative model by matching the spatial distribution of employment and wages. We measure private-sector employment and average wages by city in 2015 using administrative data from the Unemployment Fund Administrator (henceforth AFC). The AFC is a regulated private entity that manages unemployment insurance contributions made jointly by formal private-sector workers and their employers.[6]

We use data on travel times between cities to estimate transport costs. We draw these data from the Google Maps API.

3 The spatial propagation of local deposit shocks

We begin by studying empirically the elasticity of bank lending and pricing decisions with respect to deposits. These elasticities govern how deposit shocks propagate through bank networks and affect credit supply across space. We provide direct evidence on these elasticities using an instrumental variables strategy that exploits exogenous variation in commodity prices.

3.1 Empirical Strategy

We exploit variation in the world price of salmon, which disproportionately affects deposits in southern coastal cities with salmon production. We construct bank-level deposit shocks by interacting movements in salmon prices with banks' exposure to fishing cities. Our estimating equation is

$$\log L_{nbt} = \beta_1 \log D_{bt} + \beta_2 \log L_{nb,t-4} + \gamma_{nb} + \gamma_{nt} + \varepsilon_{nbt}, \tag{1}$$

where L_{nbt} is the outstanding stock of loans held by bank b in city n at month t, and D_{bt} is the aggregate stock of deposits held by bank b. We include city-bank fixed effects γ_{nb} to absorb time-invariant differences across city-bank pairs, month-city fixed effects γ_{nt} to control for aggregate local shocks, and a four-month lag of the dependent variable to account for persistence in lending relationships. We weight observations by the lagged value of loans. The coefficient of interest, β_1, captures the elasticity of lending with respect to

[6] These contributions are a fixed share of monthly wages, capped at approximately USD 5,000. We impose two additional filters on the sample. We restrict the sample to firms that appear in the Firms' Directory used by Chile's National Accounts and that employ, on average, at least three workers throughout the sample period. The final dataset includes 160,482 firms.

deposits.

The error term in equation (1) may be correlated with deposits if unobserved bank-level factors jointly affect a bank's ability to attract deposits and extend credit. We address this concern using two instruments. The first is lagged deposits $D_{b,t-4}$, which captures persistent differences in bank funding that are predetermined relative to current lending conditions. The second exploits variation in world salmon prices interacted with banks' exposure to fishing cities,

$$Z_{bt} = \underbrace{\sum_n \omega^D_{nb} \cdot \alpha^{\mathcal{F}}_n}_{\text{Exposure to fishing cities}} \times p^{\text{salmon}}_{t-3}, \tag{2}$$

where $\omega^D_{nb} \equiv D_{nb,2011} / \sum_{n'} D_{n'b,2011}$ is the share of bank b's 2011 deposits originating in city n, $\alpha^{\mathcal{F}}_n \equiv \ell^{\mathcal{F}}_n / \ell_n$ is the local employment share in fishing industries, and p^{salmon}_{t-3} is the world price of salmon lagged one quarter. Measuring banks' geographic exposure in 2011, before the start of our sample, alleviates concerns that banks endogenously enter markets in response to salmon price movements.

The salmon industry is well-suited for constructing localized deposit shocks in Chile. Salmon prices exhibited substantial variation during 2012–2018, the industry accounts for nearly 10 percent of non-copper exports, and salmon firms are headquartered locally in southern Chile, so that profits flow to local bank branches rather than to corporate offices in Santiago.[7]

The exclusion restriction requires that salmon prices and lagged deposits affect lending only through their effect on bank-level deposits. To address the plausibility of the first condition, we exclude all cities with any fishing employment from the estimation sample. To address the second condition, we include lagged loans in the estimating equation.

We also examine how deposit shocks affect loan pricing using administrative data on individual loans. We estimate

$$\log(1 + i^b_{\ell ft}) = \alpha_1 \log D_{bt} + \gamma_{nb} + \gamma_{nt} + \alpha_2 X_{ft} + \alpha_4 X_{\ell t} + \epsilon^b_{\ell ft}, \tag{3}$$

where $i^b_{\ell ft}$ is the interest rate charged by bank b on loan ℓ to firm f located in city n during month t. We restrict attention to single-city firms, for which the interest rate is arguably shaped by local conditions. We control for time fixed effects, firm characteristics X_{ft}, including employment and risk ratings, and loan characteristics $X_{\ell t}$, including amount, maturity, type, and expected loss. City-bank fixed effects capture persistent differences in local competition, and city-month fixed effects absorb changes in local economic

[7] Figure 14 in Appendix A.6 plots the evolution of world salmon prices. Figure 15 displays the geographic concentration of the fishing industry.

conditions. We instrument for deposits using the same strategy as above and weight observations by loan amount.[8]

3.2 Results

The first column in Table 2 presents the first-stage regression results of log deposits on both instruments. Lagged deposits are a strong predictor of current deposits, and banks with greater exposure to fishing cities experience significantly larger deposit inflows when salmon prices rise. The joint first-stage F-statistic is around 38, above conventional thresholds.

Our baseline estimate of the elasticity of lending with respect to deposits, shown in the column (1) of Table 2, is 0.11. This estimate is statistically significant and similar in magnitude to the elasticity of 0.16 at the city level estimated by Bustos et al. (2020) using the adoption of genetically engineered soy in Brazil as an agricultural productivity shock. Becker (2007) and Gilje et al. (2016) find positive effects on lending and the number of establishments and mortgage loans, but their results are not directly comparable to our elasticity.

In the following columns we study heterogeneous responses across cities. Column (2) of Table 2 shows that lending responses do not attenuate with distance from fishing cities. We compute each city's average travel time to fishing cities and interact this measure with instrumented deposits; the interaction term is small and statistically insignificant. This absence of distance decay contrasts with the spatial frictions typically observed for trade and migration flows, and suggests that banks' internal capital markets effectively integrate distant locations. Moreover, this result supports the exclusion restriction for our instrument in the sample of non-fishing cities.

Lending responses are concentrated in cities where banks hold small market shares. Column (3) shows that the interaction between instrumented deposits and bank market share is negative and highly significant. This pattern is consistent with oligopolistic competition between local branches: banks with larger market shares face less elastic residual demand, leading them to absorb part of the cost reduction into higher markups rather than expanding lending.

Column (5) shows our baseline estimate of the effect of deposit shocks on interest rates. The estimated elasticity of -0.008 implies that a 1% increase in deposits lowers interest rates by approximately 0.8 basis points. Consistent with the lending results, interest rate pass-through does not depend on distance but is weaker in cities where banks hold large market shares, reinforcing the role of endogenous markup responses

[8] This specification uses the subset of banks that issued loans in every year during 2012–2018 and appear in both the deposit and loan-level administrative data, leaving us with six banks, all among the largest in the country.

Table 2: The Effect of Deposit Shocks on Lending and Interest Rates

		Loans				*Interest Rates*			
	First Stage	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Deposits		0.111^{**}	0.092^{*}	0.172^{***}	0.162^{***}	-0.008^{***}	-0.007^{**}	-0.009^{***}	-0.012^{***}
		(0.049)	(0.054)	(0.038)	(0.038)	(0.001)	(0.003)	(0.001)	(0.001)
× Distance			0.014				−0.0009		
			(0.017)				(0.0023)		
× Mkt Share				-1.518^{***}	-1.457^{***}			0.021^{***}	0.031^{***}
				(0.210)	(0.210)			(0.007)	(0.007)
× D-L Ratio					0.004^{***}				-0.0958
					(0.002)				(0.514)
× Mkt Share × D-L Ratio					-0.001^{***}				−0.008
					(0.000)				(0.468)
Loans_{t-4}		0.684^{***}	0.684^{***}	0.517^{***}	0.517^{***}				
		(0.029)	(0.029)	(0.037)	(0.036)				
Deposits_{t-4}	1.213^{***}								
	(0.023)								
Instrument Z_{bt}	173.1^{***}								
	(56.0)								
City × Bank FE	–	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City × Month FE	–	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Loan Controls	–	–	–	–	–	Yes	Yes	Yes	Yes
Observations	1,480	28,439	28,439	28,127	28,127	257,495	236,782	257,495	231,498
Banks	21	15	15	15	15	6	6	6	6
First-stage F	37.78								

Notes: This table reports estimates of the effect of deposit shocks on bank lending (columns 1–4) and interest rates (columns 5–8). Columns 1-8 are estimated by 2SLS using the instrument defined in equation (2). The sample excludes cities with any employment in fishing industries. Distance is average travel time to fishing cities, in thousand kms. Market share is bank b's share of local lending in city n. D-L Ratio is the local deposit-to-loan ratio divided by 1000. Observations in columns 1–4 are weighted by total bank-level loan volume. Standard errors clustered at the bank-month level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

that depend on banks' local market shares.

Columns (4) and (8) of Table 2 explore additional heterogeneity by interacting deposit shocks, market share, and the local deposit-to-loan ratio. We expand on why this ratio matters for quantity and price responses after introducing the model in Section 4, but the broad intuition is as follows. In cities where deposits are large relative to loans, borrowers have a viable alternative to bank lending for transferring resources across periods; they can reduce investment and save more through deposits instead. This outside option makes loan demand more elastic and limits banks' ability to exercise market power. Consistent with this logic, we find that loans respond by more in cities with higher deposit-to-loan ratios. The sign of the effect on interest rates aligns with this interpretation, although it is not statistically significant.

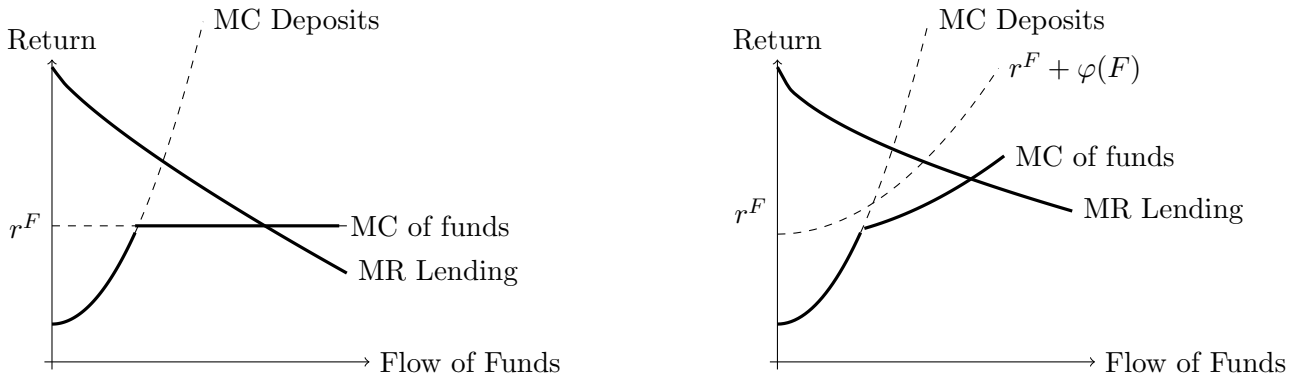
We study the robustness of our results to changing the structure of fixed effects, clustering at different levels, and using new loans rather than the stock of loans as the outcome variable. We discuss these extensions in Appendix A.7. Our main estimate of the elasticity of loan demand to deposit shocks remains robust across specifications.

We conclude by discussing banks' substitutability between deposits and other funding sources at a conceptual level in light of our empirical results. This mechanism will be central to our treatment of the interbank market in the rest of our analysis.

3.3 Banks' substitutability between deposits and other sources of funding

Our empirical results suggest that deposits and other sources of funding are not perfect substitutes. This argument can be illustrated by focusing on the two sides of the balance sheet, and distinguishing between two different sources of funds, deposits and an alternative, which we label wholesale funds in this discussion. With market power in deposits, banks need to pay more in order to attract more deposits. With market power in lending, the marginal revenue is decreasing in total loans issued. Figure 2 illustrates this basic setting and banks' responses to a shock to deposits in partial equilibrium.

In both panels, the size of banks' balance sheet is determined by equating the marginal revenue of lending with the marginal cost of funds from either retail deposits or wholesale funding. In panel (a), we assume the marginal cost in the wholesale market is constant at r^F. A positive shock to deposits (captured by an outward shift of the MC deposits curve) alters the funding composition but not the size of the balance sheet. In panel (b), where we assume the marginal wholesale costs are increasing in volume, a positive shock to deposits reduces funding costs at the margin, enabling higher lending and lower interest rates on loans. Our empirical results align with this latter case.



(a) Constant wholesale cost r^F

(b) Increasing wholesale cost $r^F + \varphi(F)$

Figure 2: Balance sheet effects of deposit shocks

As illustrated in panel (b), the effects of deposit shocks on lending and interest rates depend on two features of the environment. First, the slope of the wholesale cost curve determines how much a shift in the funding mix reduces marginal costs. Steeper wholesale costs imply that banks rely more on deposits before the shock and that the marginal cost reduction from additional deposits is larger. Second, the slope of the marginal revenue curve governs how much lending expands in response to a given reduction in marginal costs. Flatter marginal revenue—corresponding to more elastic loan demand—implies larger quantity responses and smaller price responses. Following this intuition, we use the results in Table 2 as targets to calibrate the degree of interbank frictions and the elasticity of loan demand in the quantitative version of the model.

Why are deposits and wholesale funding not perfect substitutes? A large literature argues that deposits are banks' preferred funding source due to their low cost and stability (Kashyap et al., 2002; Hanson et al., 2015). Deposits provide liquidity services to customers and are often insured, which enhances their resilience relative to market-based funding. In Chile, as in many countries, deposits are insured up to a legal threshold, further strengthening this channel. In our quantitative model, we capture these frictions parsimoniously by assuming that banks face non-pecuniary costs when borrowing on the wholesale market. Following Oberfield et al. (2024), we assume this cost to be increasing and convex in the volume borrowed.[9]

In our model we will assume that the wholesale market is an interbank market in which banks lend funds to each other. As we showed in Section 2, this approximates the banking sector in Chile where the aggregate ratio of loans to deposits is close to one. Integrating the mechanisms we study in this paper with alternative sources of funds for banks in the wholesale market is an interesting avenue in which to extend our analysis.

[9] One possible interpretation for this functional form assumption is debt overhang costs associated with wholesale funding in a risky environment, as in Andersen et al. (2019).

4 Model

We build a model that can match our empirical results quantitatively and be used to study the effects of the geographic distribution of bank branches on productivity and welfare, as well as specific policies aimed at improving financial linkages between cities. We borrow the basic structure of the model from Kleinman et al. (2023), which includes trade, migration, and endogenous investment in physical capital. Tractability comes from assuming mobile, hand-to-mouth workers and immobile capitalists. We embed the network of bank branches (which we take as given) into this structure for the rest of the economy. In line with our empirical results and studies at the intersection of industrial organization and finance, we assume that credit markets are local and banks have market power when setting local interest rates (Aguirregabiria et al., 2025).

4.1 Setup

The economy consists of N cities, indexed by n, and B banks, indexed by b. Time is discrete. There are three types of agents: workers, capitalists, and bank owners. Workers are homogeneous, do not save or borrow and are freely mobile across cities. Capitalists are immobile and reside permanently in one city, where they own the local physical capital. Capitalists rely on local bank branches for their saving and borrowing decisions. We denote the set of banks with branch presence in city n as $\mathcal{B}^n$.

We begin by specifying local production technologies and workers' migration decision. We then turn to the problem of capitalists, deriving their supply of savings and demand for loans. Finally, we introduce bank owners, their objective function, and constraints when setting interest rates.

4.1.1 Production and trade

Each location produces a differentiated good. The representative firm in location n hires labor, ℓ_{nt}, and capital, k_{nt}, from workers and capitalists, respectively, and makes production decisions in a perfectly competitive environment. The firm produces according to a Cobb-Douglas technology given by

$$y_{nt} = z_n \left(\frac{\ell_{nt}}{\mu}\right)^{\mu} \left(\frac{k_{nt}}{1-\mu}\right)^{1-\mu},$$

where z_n denotes productivity.

Trade is costly. For one unit to arrive in location n, $\tau_{ni} \geq 1$ units must be shipped from location i. The price of a good of variety i for a consumer located in n is given by

$$p_{nit} = \tau_{ni} p_{iit} = \frac{\tau_{ni} w_{it}^{\mu} r_{it}^{1-\mu}}{z_i},$$

where p_{iit} denotes the free-on-board price for the good produced in city i.

4.1.2 Workers

There is a unit mass of identical and infinitely-lived hand-to-mouth workers. The problem of a worker located in city n is as follows. First, she decides how much to consume of each of the N goods in the economy, aggregating goods from all origins with a constant elasticity of substitution,

$$C_{nt}^{w} = \left(\sum_{i=1}^{N} (c_{it}^{w})^{\frac{\sigma_c - 1}{\sigma_c}} \right)^{\frac{\sigma_c}{\sigma_c - 1}}. \tag{4}$$

The consumption price index in city n, P_{nt}, and the fraction of expenditure of city n in goods from city i, π_{nit}, are

$$P_{nt} \equiv \left(\sum_{i} (\tau_{ni} p_{iit})^{1-\sigma_c} \right)^{\frac{1}{1-\sigma_c}} \quad \text{and} \quad \pi_{nit} = \left(\frac{\tau_{ni} p_{iit}}{P_{nt}} \right)^{1-\sigma_c}. \tag{5}$$

The budget constraint of a worker is given by

$$P_{nt} C_{nt}^{w} = w_{nt}(1 - \tau)$$

where τ is a labor income tax. While the tax is zero in our baseline scenario, it will play a role in the policies we study in Section 6. After consuming in period t, the worker faces idiosyncratic utility shocks of moving to each destination city d, ϵ_{dt}, and makes her moving decision at the end of the period. Given our focus on steady state outcomes, we assume there are no migration costs.

All things considered, workers' value of living in city n at t combines an amenity value b_n, consumption utility, and the continuation value of moving:

$$v_{nt}^{w} = \log(b_n C_{nt}^{w}) + \max_{d} \{ \beta \mathbb{E}_t [v_{dt+1}^{w}] + \rho \epsilon_{dt} \}. \tag{6}$$

We assume that idiosyncratic shocks ϵ are drawn from an extreme value distribution, $F(\epsilon) = \exp(-\exp(-(\epsilon - \bar{\gamma}))$. The parameter ρ captures the relative importance of idiosyncratic reasons for migration that are not captured by amenities or real income in a city. The expectation is taken with respect to future realizations

of idiosyncratic shocks ϵ_{dt+1}.

4.1.3 Capitalists: loan demand and investment

There is one infinitely-lived immobile representative capitalist per city. The capitalist owns the local stock of physical capital and rents it to producers. To transfer resources inter-temporally, the capitalist can either invest in physical capital or save using deposits at local bank branches. Both deposits and loans are one-period, risk-free claims. In what follows, we characterize the capitalist's demand for loans and supply of deposits separately, as each plays a distinct role in the model.

To invest in physical capital, the capitalist must borrow from local banks. In practice, banks specialize in lending to firms of different sizes, sectors, or exporting status. We capture this heterogeneity by assuming that loans from different banks are imperfect substitutes from the perspective of the borrower. Specifically, one unit of investment good is produced by combining real borrowed amounts from different banks according to a CES aggregator,

$$i_{nt} = \left[\sum_{b \in \mathcal{B}_n} \left(\gamma_n^b \frac{L_{nt+1}^b}{P_{nt}} \right)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \tag{7}$$

where L_{nt+1}^b denotes loans issued in period t and maturing at $t+1$, and P_{nt} is the consumption price index from equation (5). The parameters γ_n^b capture the fit between bank b and the activities conducted in city n. The elasticity of substitution between banks, σ, governs how easily the capitalist can redirect borrowing across banks and is therefore a key determinant of banks' local market power.

The capitalist chooses how much to borrow from each bank to minimize the cost of financing a given level of investment:

$$\mathcal{L}_{nt}(i_{nt}) = \min_{\{L_{nt+1}^b\}_b} \sum_{b \in \mathcal{B}_n} L_{nt+1}^b (1 + r_{nt+1}^b) \quad \text{s.t.} \quad \text{equation (7)}.$$

The solution yields a demand function for loans from bank b that takes the CES form:

$$\frac{L_{nt+1}^b}{P_{nt}} = (\gamma_n^b)^{\sigma-1} \left(\frac{R_{nt+1}}{1 + r_{nt+1}^b} \right)^{\sigma} i_{nt}, \tag{8}$$

where

$$R_{nt+1} \equiv \left[\sum_{b \in \mathcal{B}_n} \left(\frac{1 + r_{nt+1}^b}{\gamma_n^b} \right)^{1-\sigma} \right]^{\frac{1}{1-\sigma}} \tag{9}$$

is the loan price index—the effective cost of one unit of investment financing. Demand for loans from bank b is decreasing in r_{nt+1}^b relative to the price index R_{nt+1}, with the elasticity governed by σ. The total cost

of financing investment is

$$\mathcal{L}_{nt}(i_{nt}) = i_{nt} R_{nt+1} P_{nt}. \tag{10}$$

4.1.4 Capitalists: deposits, consumption, and the full problem

Having characterized how the capitalist finances investment, we now turn to her broader inter-temporal problem. Following the finance literature, we assume that capitalists derive utility from both consumption and the liquidity services provided by deposits (Drechsler et al., 2017; Morelli et al., 2024). The parameter α controls the weight on liquidity benefits. Deposits from different banks are aggregated with an elasticity of substitution η, with parameters κ_n^b capturing differences in the utility of deposits across banks associated, for example, with a bank having more branches in the city.

The capitalist's problem is[10]

$$\max_{\{C_{nt}^c, D_{nt+1}^b, k_{nt+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \left[\log C_{nt}^c + \alpha \log D_{nt+1}\right], \tag{11}$$

where

$$D_{nt+1} = \left[\sum_b (\kappa_n^b D_{nt+1}^b)^{1-\frac{1}{\eta}}\right]^{\frac{\eta}{\eta-1}},$$

subject to the budget constraint

$$P_{nt} C_{nt}^c + \sum_b D_{nt+1}^b + i_{nt-1} R_{nt} P_{nt-1} = \hat{r}_{nt} k_{nt} + \sum_b D_{nt}^b (1 + \tilde{r}_{nt}^b) + T_{nt}^c \tag{12}$$

and the capital accumulation equation $k_{nt+1} = k_{nt}(1-\delta) + i_{nt}$.

The budget constraint has a natural interpretation: income on the right-hand side comes from renting out capital at rate $\hat{r}_{nt}$, the return on deposits maturing at t, and a government transfer T_{nt}^c specified below. This income finances consumption, new deposits, and the repayment of loans that were taken out at $t-1$ to finance investment.

The first-order conditions yield a demand for deposits from bank b that mirrors the CES structure of loan demand:

$$D_{nt+1}^b = (\kappa_n^b)^{\eta-1} \left(\frac{Q_{nt+1}}{q_{nt+1}^b}\right)^{\eta} D_{nt+1}. \tag{13}$$

[10] We assume that the intra-temporal consumption problem of a capitalist is equivalent to the workers' with the same elasticity of substitution across goods from different locations.

Here q^b_{nt+1} is the effective price of a deposit with bank b, defined as

$$q^b_{nt+1} \equiv 1 - \frac{1 + \tilde{r}^b_{nt+1}}{(1-\delta)R_{nt+1}P_{nt}/(R_{nt}P_{nt-1} - \hat{r}_{nt})}, \tag{14}$$

and Q_{nt+1} is the corresponding deposit price index.

The expression for q^b_{nt+1} captures the key economic trade-off facing the capitalist. The numerator of the adjustment factor is the gross return on deposits. The denominator is the gross return on physical capital investment, derived from the Euler equation. A deposit is costly to the extent that its pecuniary return falls short of the return the capitalist could earn by investing in physical capital instead. When deposit rates are high, or when investment returns are low, q^b_{nt+1} is low and deposits become relatively more attractive.

This connection between q^b_{nt+1} and the return on investment plays an important role in the rest of our analysis, where we show that capitalist's demand for deposits shapes the elasticity of *loan* demand faced by banks and, therefore, the markups they charge on loans.

Combining the first-order conditions for deposits and consumption yields closed-form solutions for aggregate deposits and capitalist consumption:

$$D_{nt+1} = \frac{\alpha M_{nt}}{Q_{nt+1} + \alpha Q^{\eta}_{nt+1}\tilde{Q}_{nt+1}}, \tag{15}$$

$$P_{nt}C^c_{nt} = \frac{Q_{nt+1}M_{nt}}{Q_{nt+1} + \alpha Q^{\eta}_{nt+1}\tilde{Q}_{nt+1}}, \tag{16}$$

where $M_{nt} \equiv \hat{r}_{nt}k_{nt} + \sum_b (1+\tilde{r}^b_{nt})D^b_{nt} - i_{nt-1}R_{nt}P_{nt-1}$ is the capitalist's income and $\tilde{Q}_{nt+1}$ is an alternative index across local q^b_{nt+1}, defined in Appendix Section B.2.

The key demand functions stemming from capitalists' behavior are

$$L^b_{nt+1} = (\gamma^b_n)^{\sigma-1}\left(\frac{R_{nt+1}}{1+r^b_{nt+1}}\right)^{\sigma} i_{nt}P_{nt}, \tag{17}$$

$$\text{and } D^b_{nt+1} = (\kappa^b_n)^{\eta-1}\left(\frac{Q_{nt+1}}{q^b_{nt+1}}\right)^{\eta} \frac{\alpha M_{nt}}{Q_{nt+1} + \alpha Q^{\eta}_{nt+1}\tilde{Q}_{nt+1}}. \tag{18}$$

which banks' will take as given. Loan demand from bank b is decreasing in the bank's interest rate r^b_{nt+1}; deposit supply to bank b is increasing in the deposit rate $\tilde{r}^b_{nt+1}$ (which corresponds to a decrease in q^b_{nt+1}). We now turn to how banks set these rates.

4.1.5 Banks

The owner of bank b operates branches in a set of cities denoted by $\mathcal{C}^b$. The cash flow of bank b at time t is

$$\Pi_t^b \equiv \left\{ \sum_{n\in\mathcal{C}^b} \overbrace{L_{nt}^b(1+r_{nt}^b)(1-\tau_n^b)+D_{nt+1}^b}^{\text{Retail inflow}} - \overbrace{L_{nt+1}^b - D_{nt}^b(1+\tilde{r}_{nt}^b)}^{\text{Retail outflow}} \right\} + F_{t+1}^b - (1+r_t^F)F_t^b - T_t^b.$$

Retail inflows at time t consist of loans maturing at t and new deposits issued at t in all cities where the bank has branches. Retail outflows consist of loans issued at t and deposits maturing at t. The term τ_n^b represents city–bank–specific taxes on loans ($\tau_n^b < 0$ for subsidies). These taxes are zero in our baseline scenario, but they play a role in the policy analysis in Section 6, where we study policies that correct market power. The position of each bank in the interbank market is denoted by F_{t+1}^b. All banks have access to the same interbank interest rate r_t^F, and a positive value of F_{t+1}^b indicates that the bank borrows from other banks. The term T_t^b is a bank-specific lump-sum tax, defined below.

Bank owners choose city-specific nominal interest rates on loans r_{nt+1}^b and the cost of deposits q_{nt+1}^b to maximize the discounted sum of cash flows net of a non-pecuniary cost of tapping into the interbank market, which captures the forces discussed in subsection 3.3.[11] Non-pecuniary costs are assumed to be increasing in the amount borrowed or lent in the interbank market, as in Oberfield et al. (2024). The parameter ϕ governs the elasticity of non-pecuniary costs with respect to volume. Finally, banks must satisfy the balance sheet constraint, equation (20).

The problem of a bank owner is therefore

$$\max_{\{\{r_{nt}^b, q_{nt}^b\}, F_t^b\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \left\{ \Pi_t^b - \big(\exp(\phi\omega^b) - 1\big)(1+r_t^F)F_t^b \right\} \tag{19}$$

$$\underline{\text{s.t.}}\ [\mu_t^b] \sum_{n\in\mathcal{C}^b} L_{nt+1}^b = \sum_{n\in\mathcal{C}^b} D_{nt+1}^b + F_{t+1}^b \quad \forall t, \tag{20}$$

$$\text{equation (9)},\ \text{equation (17)},\ \text{equation (18)} \quad \forall t,\ \forall n \in \mathcal{C}^b,$$

where we defined reliance on the interbank market as

$$\omega^b \equiv \frac{F_t^b}{\sum_{n\in\mathcal{C}^b} D_{nt}^b}.$$

We assume oligopolistic competition in the loan market and monopolistic competition in the market for

[11] We write the bank's problem in terms of the deposit cost q_{nt+1} instead of $\tilde{r}_{nt+1}$ for simplicity; the interest rate can be recovered from equation (14).

deposits. That is, the bank owner takes the demand for deposits and loans given by equation (18) and equation (17) as given and internalizes its own effect on the interest rate index R_{nt}, but not on Q_{nt} or $\tilde{Q}_{nt}$.[12]

From the first-order conditions of this maximization problem, the marginal cost of issuing loans for bank b is:

$$\mathcal{MC}_t^b \equiv \left(\tfrac{1}{\beta} + \mu_t^b\right) = \exp(\phi\omega^b)(1 + r_{t+1}^F)\big(1 + \phi\omega^b\big). \tag{21}$$

From the perspective of a bank, the marginal cost of issuing a loan includes the dollar the bank must give up today in exchange for a dollar tomorrow, in addition to the value of balance sheet space, captured by μ_t^b. The latter depends on how much the bank is currently tapping into the interbank market, as shown in the last expression in equation (21).

Optimal local interest rates satisfy

$$(1 + r_{nt+1}^{b*})(1 - \tau_n^b) = \frac{\varepsilon_{nt}^{Lb}}{\varepsilon_{nt}^{Lb} - 1}\,\mathcal{MC}_t^b, \tag{22}$$

$$q_{nt+1}^b = -\frac{\eta}{\eta - 1}\beta\left\{\exp(\phi\omega^b)(1 + r_{t+1}^F)\phi(\omega^b)^2 + \mathcal{MC}_t^b - \frac{1}{\beta}\right\}. \tag{23}$$

where $\varepsilon_{nt}^{Lb} \equiv -\frac{\partial L_{nt}^b}{\partial r_{nt}^b}\frac{(1+r_{nt}^b)}{L_{nt}^b}$ is the demand elasticity of loans. Equation (22) shows how loan markups vary across cities depending on the local sensitivity of loan demand to interest rates. The local elasticity faced by a particular bank is

$$\varepsilon_{nt}^{Lb} \equiv -\frac{\partial L_{nt}^b}{\partial r_{nt}^b}\frac{(1 + r_{nt}^b)}{L_{nt}^b} = \sigma\big(1 - s_{nt+1}^b\big) + s_{nt+1}^b\varepsilon_{nt+1}^i, \tag{24}$$

$$\text{where } \varepsilon_n^{i,R} \equiv -\frac{\partial i_{nt}}{\partial R_{nt+1}}\frac{R_{nt+1}}{i_{nt}} \overbrace{=}^{\text{at the steady state}} \frac{1}{\beta(1-\delta)}\left[1 + \frac{D_n Q_n}{\alpha i_n R_n P_n}\right]. \tag{25}$$

and $s_{nt+1}^b \equiv \frac{(1+r_{nt+1}^b)L_{nt+1}^b}{i_{nt}R_{nt+1}P_{nt}}$ is bank b's local revenue share.

As in Atkeson and Burstein (2008), the relevant elasticity is a revenue-share-weighted average of the local elasticity of substitution between banks, σ, and the city-level aggregate elasticity of investment with respect to the price index, $\varepsilon_n^{i,R}$. We discuss each of these objects in more detail in Section 4.3 below. Similar expressions feature in the models of loan demand in Herreno (2023) and Altavilla et al. (2022). The main feature of our setting is that the aggregate elasticity of investment $\epsilon_n^{i,R}$ is linked to the local availability of deposits.

Equation (23) shows that markdowns on deposits are constant in the model, which follows from our

[12] We exclude oligopolistic competition in deposits to keep the analysis focused on loan rates and because we do not have detailed data on deposit rates to include in our empirical analysis, but it could be easily incorporated into our framework.

assumption of monopolistic competition in the deposit market. The right-hand side of equation (23) includes an additional term because capturing deposits reduces non-pecuniary costs.[13]

In Section 3 we documented that, following a shock to its deposit base, a bank issues more loans and lowers its interest rates, and we used a simple partial equilibrium model to explain these results in subsection 3.3. The quantitative model captures the same intuition: an increase in deposits lowers marginal costs in the right-hand side of equation (21), which translates into lower interest rates through equation (22) and leads to higher lending through the loan demand function equation (17).

4.1.6 Fiscal policy

The government collects taxes and transfers the revenue back into the economy. In the baseline scenario, the government levies taxes on banks and rebates the revenue to capitalists, both lump-sum from the perspective of banks and capitalists. Taxes on bank b are

$$T_t^b = \sum_{n \in \mathcal{C}^b} L_{nt}^b (r_{nt}^b (1 - \tau_n^b) - \tau_n^b) - D_{nt}^b \tilde{r}_{nt}^b - r_t^F F_t. \tag{26}$$

The taxes defined in equation (26) are such that after-tax bank cash flows are zero at the steady state, which makes the geographic location of bank owners irrelevant. The government uses these funds to finance lump-sum transfers to capitalists,

$$T_{nt}^c = \sum_{b \in \mathcal{B}^n} L_{nt}^b r_{nt}^b - D_{nt}^b \tilde{r}_{nt}^b. \tag{27}$$

With the transfers defined in equation (27), capitalist's net interest losses (or gains) from interacting with their local branches are undone by the government transfers. After defining the steady state, we show that government finances are balanced.

Counterfactual analysis. To study the role of market power, we implement policies that undo markups into our baseline model. These city-bank specific subsidies τ_n^b are fully financed by a labor-income tax τ, which adjusts endogenously to satisfy the government's budget balance condition

$$\tau \sum_n w_n \ell_n = -\sum_{b=1}^{B} \sum_{n \in \mathcal{C}^b} L_n^b (1 + r_n^b) \tau_n^b. \tag{28}$$

As we show below, proportional taxes on labor income do not distort workers' moving decisions, and

[13] For analyses of market power on the deposit side, see Drechsler et al. (2017) and Albertazzi et al. (2024).

therefore provide a useful tool to undo the distortions coming from market power without incorporating other distortions. Assuming that these policies are fully financed by taxing workers, naturally, is highly demanding on the effect they can have on workers' welfare.

4.2 Steady state

Given a vector of productivity and amenity values, $\{z_n, b_n\}_{n\in N}$, the set of cities in which each bank is present, $\{\mathcal{C}^b\}_{b\in B}$ and fiscal policy $\tau, \{T^b\}_{b\in B}, \{T_n, \{\tau_n^b\}_{b\in\mathcal{B}^n}\}_{n=1}^N$, a steady state consists of a vector of prices $r^F, \{w_n, p_n, \{r_n^b, \tilde{r}_n^b\}_{b\in B}\}_{n\in N}$, and quantities $\{F^b\}_{b\in B}, \{\ell_n, k_n, i_n, y_n, C_n^w, C_n^c, k_n, \{L_n^b, D_n^b\}_{b\in B}\}_{n\in N}$, that satisfy: (i) optimality for consumption shares, equation (5); (ii) the labor market clearing condition:[14]

$$\ell_n = \frac{\left(\frac{b_n w_n (1-\tau)}{P_n}\right)^{\frac{\beta}{\rho}}}{\sum_{i=1}^N \left(\frac{b_i w_i (1-\tau)}{P_i}\right)^{\frac{\beta}{\rho}}} \quad \forall n, \tag{29}$$

where ℓ_n is labor demand from local firms; (iii) capitalist's consumption, borrowing, and saving optimality conditions, equation (16), equation (18) and equation (17); (iv) optimality conditions from the bank owner's problem, equation (21), equation (22) and equation (23); (v) market clearing for final goods

$$w_n \ell_n + \hat{r}_n k_n = \sum_{i=1}^N \pi_{ni} \left(P_i C_i^w + P_i C_i^c + \sum_{b\in\mathcal{B}^i} L_i^b \right) \quad \forall n, \tag{30}$$

where consumption of city n goods comes from workers and capitalists nationally; (vi) market clearing in the interbank market,

$$\sum_b F^b = 0; \tag{31}$$

(vii) the definition of bank taxes and capitalist's transfers, equation (26) and equation (27); and $(viii)$ all variables are time invariant.

4.2.1 Fiscal policy at the steady state

Bank profits are fully taxed at the steady state: Bank b's cash flows are

[14] See Section B.1 for a derivation.

$$\Pi_t^b = \Big\{ \sum_n L_{nt}^b (1 + r_{nt}^b)(1 - \tau_n^b) + D_{nt+1}^b - L_{nt+1}^b - D_{nt}^b (1 + \tilde{r}_{nt}^b) \Big\} + F_{t+1}^b - (1 + r_t^F) F_t^b - T_t^b.$$

At a steady state in which $L_n^b = L_{nt}^b, D_n^b = D_{nt}^b$, and $F^b = F_t^b$ for all t, and using equation (26),

$$\Pi^b = \Big\{ \sum_n L_n^b (r_n^b (1 - \tau_n^b) - \tau_n^b) - D_n^b \tilde{r}_n^b \Big\} - r^F F^b - T^b = 0.$$

Budget balance: The budget constraint of the government in the baseline case, $\tau_n^b = 0 \ \forall n, b$ is satisfied if

$$\begin{aligned} \sum_{n=1}^{N} T^c &= \sum_{b=1}^{B} T^b \\ \sum_{n=1}^{N} \sum_{b \in \mathcal{B}^n} L_n^b r_n^b - D_n^b \tilde{r}_n^b &= \sum_{b=1}^{B} \sum_{n \in \mathcal{C}^b} L_n^b r_n^b - D_n^b \tilde{r}_n^b - r^F F^b \\ r^F \sum_{b=1}^{B} F^b &= 0 \end{aligned}$$

which follows from market clearing in the interbank market equation (31).

4.3 The determinants of local interest rates

Bordeu et al. (2025) document substantial dispersion in interest rates across Chilean cities, even after controlling for borrower characteristics and the identity of the lending bank, suggesting that credit markets have a strong local component. In line with this idea, our empirical results in Section 3 show that interest rate responses differ across cities depending on local market structure. In this section, we show how the model generates city-specific interest rates as an equilibrium outcome and discuss the mechanisms at play. We begin with a simplified version of the model to build intuition, then present the general case.

A simplified benchmark. Consider a version of the model in which there are no city-bank-specific matches ($\gamma_n^b = 1$ for all n, b) and no interbank frictions ($\phi = 0$). In this case, all banks face the same marginal cost, equal to the interbank rate $\mathcal{MC}^b = 1 + r^F$. By symmetry, all banks charge identical interest rates in a given city and hold equal market shares, $s_n^b = 1/B_n$, where B_n is the number of banks present.

The common interest rate in city n then satisfies

$$1 + r_n = \frac{\sigma B_n - \Delta_n}{B_n(\sigma - 1) - \Delta_n}(1 + r^F), \tag{32}$$

where $\Delta_n \equiv \sigma - \varepsilon_n^{i,R}$ is the gap between the elasticity of substitution across banks and the aggregate elasticity of investment demand with respect to the loan price index. equation (32) highlights two determinants of local interest rates: the number of competing banks and the sensitivity of local investment to borrowing costs.

The markup in equation (32) can be rewritten as $(\sigma - \Delta_n/B_n)/(\sigma - \Delta_n/B_n - 1)$. As the number of banks increases, $\Delta_n/B_n \to 0$ and the markup converges to the standard monopolistic competition benchmark, $\sigma/(\sigma - 1)$, which obtains when each bank's market share is negligible. The rate of convergence depends on Δ_n: in cities where the gap between the inner and outer elasticities is large, adding a bank has a bigger effect on markups. When $\Delta_n > 0$ (which, as we discuss below, is the empirically relevant case), it follows that

$$\frac{\partial(1 + r_n)}{\partial B_n} = \frac{-\Delta_n}{[B_n(\sigma - 1) - \Delta_n]^2}(1 + r^F) < 0$$

and more banks unambiguously lower local interest rates. In cities where Δ_n is large, local investment demand is relatively inelastic at the aggregate level, making the number of banks a more important determinant of interest rates.

The general case. In the full model, bank-specific marginal costs and city-bank matches generate asymmetric market shares, leading to the steady-state general markup expression from equation (24):[15]

$$1 + r_n^b = \underbrace{\frac{\sigma - s_n^b \Delta_n}{\sigma - s_n^b \Delta_n - 1}}_{\text{markup}} \mu^b \tag{33}$$

The markup is increasing in s_n^b whenever $\Delta_n > 0$: A bank with a larger local market share faces less elastic demand because borrowers have fewer alternatives, pushing the effective elasticity toward the lower outer elasticity $\varepsilon_n^{i,R}$.

The effective elasticity of loan demand faced by bank b in city n is given by equation (24). As in Atkeson and Burstein (2008), this is a revenue-share-weighted average of two elasticities: σ, which governs substitution between banks for a given level of investment, and $\varepsilon_n^{i,R}$, which governs how aggregate investment responds

[15] All derivations in this subsection are relegated to Appendix Section B.4.

to the overall cost of borrowing. Similar expressions feature in the models of loan demand in Herreno (2023) and Altavilla et al. (2022).

A distinguishing feature of our framework is that $\varepsilon_n^{i,R}$ is not a structural parameter but an endogenous object linked, in steady state, to the local availability of deposits:

$$\varepsilon_n^{i,R} = \frac{1}{\beta(1-\delta)}\left(1 + \frac{D_n Q_n}{\alpha i_n R_n P_n}\right). \tag{34}$$

The ratio $D_n Q_n/(\alpha i_n R_n P_n)$ compares the value of the capitalist's deposit portfolio to her investment expenditure. In cities where this ratio is high, capitalists rely more on deposits as a means of inter-temporal transfer and investment demand is more elastic, because capitalists can more easily substitute away from investment toward saving. A high deposit-to-investment ratio therefore tightens the competitive environment that banks face, limiting the role of local market shares in loans.

This expression for the outer elasticity contrasts with Atkeson and Burstein (2008), where both the inner and outer elasticities are fixed parameters of consumer preferences. In our setting, the outer elasticity responds to equilibrium conditions, creating an additional channel through which local financial conditions shape markups. We can take this prediction to the data. Columns (4) and (8) of Table 2 interact deposit shocks with the local deposit-to-loan ratio and market share. Consistent with the model, we find that loan growth is significantly higher in cities with a higher deposit-to-loan ratio, where the model predicts more elastic demand. The corresponding coefficient for interest rates has the predicted sign but is not statistically significant.

The role of interbank frictions. In the simplified model, equation (32) shows how the interbank market transmits shocks across cities: an increase in loan demand elsewhere raises r^F, crowding out lending in city n. In the full model, interbank frictions create additional heterogeneity across banks. To understand their effect on local interest rates, we take a first-order approximation around the frictionless benchmark ($\phi = 0$) and decompose the change in the loan-weighted average interest rate:[16]

$$1 + r_n(\phi) \approx (1 + r_n(0))(1 + 2\phi\omega_n) \tag{35}$$

where $\omega^b \equiv F^b/D^b$ is bank b's reliance on the interbank market (positive for net borrowers, negative for net lenders) and $\bar{\omega}_n \equiv \frac{1}{B_n}\sum_{b\in\mathcal{B}_n}\omega^b$ is the simple average across banks in city n.[17]

[16]The factor of 2 in equation equation (35) comes from the functional-form assumption on nonpecuniary costs of tapping into the interbank market.

[17]Throughout this section, F^b and D^b denote bank b's total (across all cities) interbank position and deposits, respectively.

This approximation shows that local interest rates depend on whether the banks lending in the city are connected to deposits. In cities with branches from banks that rely mostly on deposits for funding, $\omega_n \approx 0$ and local interest rates are lower. The strength of this mechanism depends on the size of interbank frictions.

Our model introduces two key ingredients relative to benchmark models at the intersection of banking and spatial economics: frictions in the interbank market and oligopolistic competition in local credit markets. Having illustrated the role of these channels theoretically, we now turn to estimating the model to assess their quantitative importance.

5 Estimation

We estimate the model by matching our empirical results in Section 3 and the spatial distribution of employment, wages, and lending in 2015. Table 3 lists the parameters we borrow from the literature as well as internally estimated parameters with their empirical counterparts.

We borrow μ, δ, β and ρ from Kleinman et al. (2023), who parameterize their model to the U.S. economy. We set the value of the elasticity of substitution across final goods to 4, which is standard in the literature, and assume that transport costs are a function of travel times, namely $\tau_{ij} = t_{ij}^{0.375}$ (Redding and Rossi-Hansberg, 2017). We borrow the deposit-elasticity of substitution across banks from Albertazzi et al. (2024), who study deposit markdowns in Europe.

We estimate the loan-elasticity of substitution across banks, interbank frictions, and the vector of productivities, amenities and city-bank matches jointly. The loan-elasticity of substitution and the parameter for interbank frictions are tightly connected to the effect of deposit shocks on lending and interest rates from Section 3. For higher values of σ, the increase in lending at the bank-level is associated with smaller reductions in interest rates. For higher values of ϕ, deposit inflows have a stronger effect on lending, as deposits constitute the main source of funds.

To replicate our empirical exercise in the model, we increase the city-bank deposit shifter κ_n^b proportionally to bank b's exposure to the fishing industry, measured as the deposit-weighted average of fishing employment shares across the cities where the bank collects deposits. This generates a bank-level deposit inflow analogous to the one induced by salmon price movements in the data: banks with a larger share of deposits originating in fishing cities experience a larger increase in their deposit base driven by a higher κ_n^b.

We solve the model under the new values of κ_n^b and compare the resulting equilibrium to the baseline. Using the model-generated data, we then replicate our empirical strategy by instrumenting for bank-level deposit growth with exposure to the shock and estimating the effects on city-bank loan volumes and interest

rates. We search over values of ϕ and σ and select the pair that matches the estimated effects of deposit shocks on quantities and prices reported in columns (1) and (5) of Table 2. As discussed in Appendix C.1, we are able to exactly match these empirical elasticities.

We estimate the vector of city-bank matches for loans and deposits, γ_n^b and κ_n^b, to match the observed values of loans and deposits in each city-bank in 2015. Using the wages observed in the data, we estimate the value of productivity $\{z_n\}$ and amenities $\{b_n\}$ in each city as those that rationalize observed labor shares and such that model-implied market clearing conditions hold. For a full description of the estimation algorithm, see Section C.1 and Section C.2.

Table 3: Estimated Parameters

A. External sources			
	Description	Value/Range	Source or Objective
μ	Capital share	0.65	Kleinman et al. (2023)
δ	Rate of depreciation	0.05	Kleinman et al. (2023)
β	Discount factor	0.95	Kleinman et al. (2023)
ρ	Such that the elasticity of migration to ϵ_d is $\frac{1}{3}$	3β	Kleinman et al. (2023)
σ_c	Elasticity of substitution (consumption)	4	Redding and Rossi-Hansberg (2017)
$\{\tau_{nj}\}_{n,j=1,..,N}$	Elasticity of trade costs to travel times t_{ij}	$t_{ij}^{0.375}$	Redding and Rossi-Hansberg (2017)
η	Elasticity of substitution (deposits)	1.6	Albertazzi et al. (2024)
B. Internally estimated			
ϕ	Cost of wholesale funding	0.004	Quantity IV results in Section 3
σ	Elasticity of substitution (loans)	14.11	Price IV results in Section 3
$\{z_n\}_{n=1}^N$	Productivity	[0.26, 0.48]	Wages
$\{b_n\}_{n=1}^N$	Amenity	[0.79, 1.16]	Employment
$\{\{\gamma_n^b\}_{b\in\mathcal{B}^n}\}_{n=1}^N$	Bank-city match	[9.19, 10.93]	Loans
$\{\{\kappa_n^b\}_{b\in\mathcal{B}^n}\}_{n=1}^N$	Bank-city match	[0.10, 1.51]	Deposits

Notes: For productivity, amenity and the bank-city matches, the range shows the 25th-75th percentile ranges.

5.1 Discussion

Elasticity of loan demand across banks. Our estimate for the elasticity of loan demand across banks is 14.11, which lies within the wide range of estimates available in the literature. Maingi (2026) estimates a range of 1.14 to 2.06 using U.S. data, while Altavilla et al. (2022) estimate values between 7 and 22 using European data. Our estimation of σ relies on matching the effects of deposit shocks on interest rates which, as explained in Section 3.3, are closely linked to the slope of loan demand. The GMM approach in Maingi

(2026) focuses instead on matching the banks' increase in lending, relative to local aggregates.

Besides the differences in the estimation approaches, several factors could explain this differences in estimates: borrowers in Chile may face lower switching costs, relationship lending may be less prevalent, or the composition of firms in our sample may differ. Given the importance of this elasticity in shaping local market power, understanding its determinants across contexts is an important avenue for future research.

Interbank frictions. In our baseline calibration, the median borrowing bank finances 30 percent of its total funding through the interbank market. Our estimate of $\phi = 0.004$ implies that such a bank behaves as if the interbank rate were 13 basis points higher than the market rate.

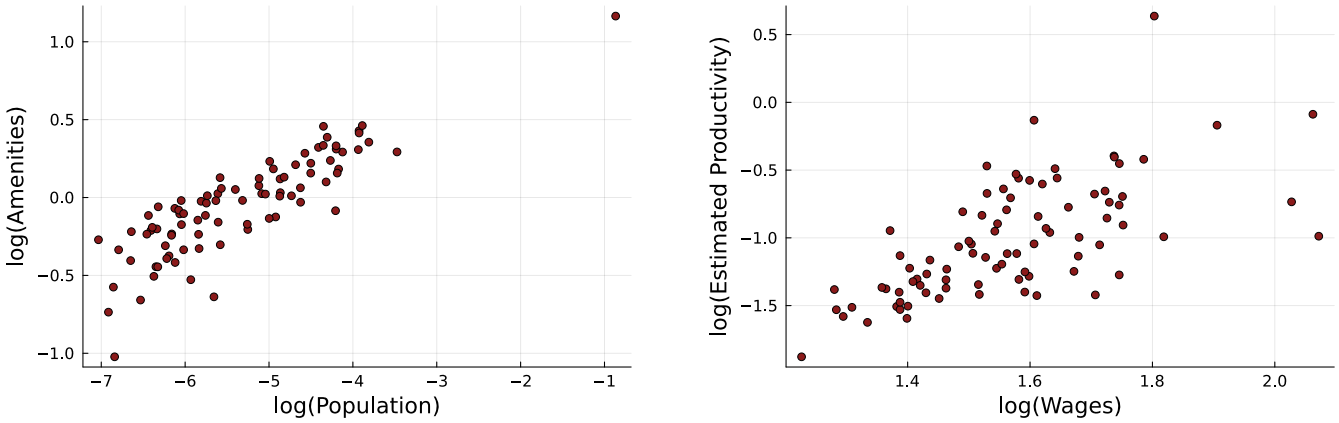
Our estimate of the frictions is not independent from our estimate of the elasticity of substitution between banks. To see this, consider a bank that receives an exogenous inflow of deposits. If demand for loans was very inelastic, the bank would prefer to lend in the interbank market rather than to retail lending, to avoid decreasing prices. To match our quantity responses in Table 2, therefore, interbank frictions would need to be substantially higher. In particular, if we had assumed a value of $\sigma = 4$ instead of $\sigma = 14.11$, our estimate of ϕ would have been 3.8 times larger.

Productivity and amenities. Figure 3a shows the estimated local amenities against employment shares. The two are tightly connected through the lens of the model. Figure 3b shows the estimated productivity values against average local wages from the data. Wages and productivity are positively related, but the relationship is not as tight because the model imposes market clearing, which introduces additional constraints on wages besides the direct effect of productivity.

City-bank matches. The full estimates of $\{\gamma_n^b\}$ are shown in Section C.2 in the Appendix. While micro-founding the origins of city-bank matches is beyond the scope of this paper, we find a positive role for the number of local branches (which may reduce the distance between clients and the bank). We estimate

$$\hat{\gamma}_n^b = \beta_0 + \beta \times \text{LogBranches}_n^b + \gamma_n + \gamma_b + \epsilon_n^b$$

using data on the number of branches in each city-bank pair in December 2015. The left-hand side includes our estimates of city-bank matches. By including city fixed effects, our results capture the effect of having a higher share of the local branches. By including bank fixed effects, our results are not mechanically capturing other qualities that differentiate banks. We estimate a positive and statistically significant coefficient on log-branches, indicating that the number of branches within cities plays a role (Appendix Table 10).



(a) Employment Data and Estimated Amenities

(b) Wage Data and Estimated Productivities

Figure 3: Estimated Residential Amenities and Productivity Parameters

6 Interbank frictions, market power, and the spatial allocation of capital

A large literature has argued that the misallocation of factors of production across establishments is a major source of aggregate productivity losses (Hsieh and Klenow, 2009).[18] Identifying the specific frictions behind differences in factor prices for different firms is a crucial first step in addressing the misallocation of resources.

In our setting, interbank frictions create dispersion in banks' cost of funds, and oligopolistic competition creates dispersion in markups across cities. Both channels raise interest rates in some cities relative to others, distorting investment and generating spatial variation in the marginal productivity of capital. Figure 4 shows the resulting distribution of MPK in our baseline calibration. In the rest of this section, we eliminate each source of misallocation in turn and quantify its contribution to aggregate productivity and welfare. Table 4 summarizes the results. Our main finding is that local market power accounts for the bulk of the distortion: equalizing markups across space raises steady-state GDP by 0.52 percent, while eliminating interbank frictions has negligible aggregate effects.

[18] See Bergquist et al. (2026) for a recent survey.

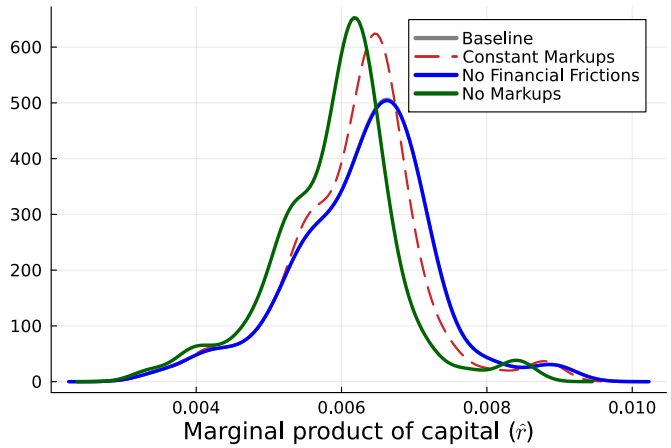


Figure 4: Spatial dispersion of the marginal productivity of capital

A growing literature discusses how to characterize welfare and productivity in spatial models (Baqaee and Burstein, 2026). Given our assumption of no migration costs, expected worker's welfare, given by

$$\overline{V}^w = \left(\sum_{n=1}^{N} \left(\frac{b_n w_n}{P_n} \right)^{\theta} \right)^{\frac{1}{\theta}}$$

equalizes across cities.[19] Given that workers are homogeneous, we focus on this object as our measure of worker's welfare.

We measure capitalist's welfare directly,

$$V_n^c = C_n^c D_n^{\alpha}.$$

and report changes in the average and median welfare across capitalists. In all counterfactual exercises, subsidies correcting markups are financed by a proportional labor income tax τ, which does not distort workers' moving decisions. This financing assumption is, naturally, demanding on the effect of each experiment on workers' welfare. We report both pre-tax and post-tax effects on workers' welfare.

No interbank frictions. We begin by eliminating the non-pecuniary costs of accessing the interbank market, setting $\phi = 0$ and recomputing the steady state. This equalizes the marginal cost of funds across banks. The aggregate effects are small, and GDP rises by 0.02%. This negligible aggregate effects masks heterogeneous responses across city-bank pairs. While equalizing the costs of funds across banks would have,

[19] Realized welfare does not, as it includes idiosyncratic shocks.

Table 4: The role of interbank frictions and market power

	No interbank frictions	No markups	Constant markups
Steady state outcomes			
Aggregate productivity	0.02%	3.55%	0.52%
Average MPK ($E[\hat{r}_n]$)	0.03%	-6.02%	-1.73%
MPK dispersion ($\text{std}(\hat{r}_n)/E[\hat{r}_n]$)	0.02%	-1.52%	-1.52%
Workers			
Welfare	0.01%	-0.02%	-0.02%
Pre-tax welfare	0.01%	3.60%	0.57%
Capitalists			
Average welfare	-0.07%	4.14%	1.01%
Median welfare	0.26%	3.86%	0.79%

all else equal, compressed spatial disparities between cities, markups respond endogenously in a way that more than offsets the direct effect. Dispersion in MPK relative to its mean in fact rises by 0.02%.

Figure 5 illustrates the mechanism, showing differences in markups and costs across city-bank pairs. Banks experiencing a reduction in marginal costs capture larger market shares and respond by increasing their markups. Conversely, banks that previously benefited from preferential access to deposits now face greater competition for funds. Their marginal costs rise, their market shares shrink, and they reduce their markups.

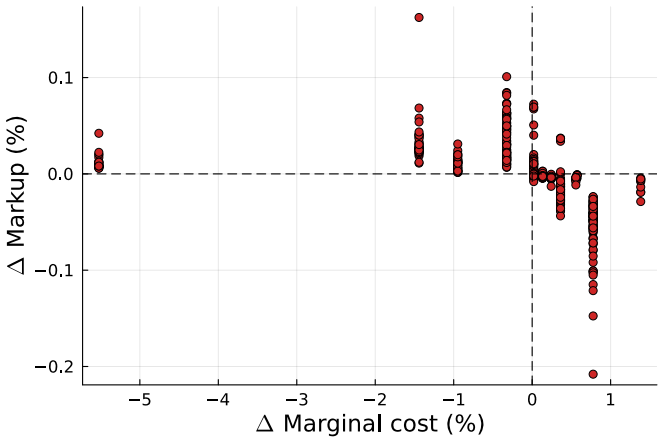


Figure 5: Markup responses to changes in marginal cost

Despite the limited aggregate gains, the spatial effects are heterogeneous, as we discussed analytically in Section 4.3. Some cities benefit from the geographic segmentation of capital markets, as they have banks with ample access to deposits. When interbank lending becomes frictionless, banks with a local presence in these cities find it easier to channel funds elsewhere. Figure 6a plots investment responses against exogenous

city productivity z_n. Cities at the bottom of the productivity distribution lose capital, while capital flows toward more productive cities. In Chile, this channel tends to benefit the least productive cities.

In terms of welfare, the effects of reducing interbank frictions are also small. Worker welfare increases by 0.01%. The median capitalist gains 0.26%, but the average capitalist loses 0.07%, reflecting the heterogeneous effects across cities.

Constant markups. We now turn to market power, which Table 4 identifies as the quantitatively dominant channel. We present two exercises. The first isolates the cost of spatial variation in markups by equalizing them across cities; the second eliminates markups entirely.

To equalize markups, we solve for the steady state of an economy in which city-bank specific subsidies replicate monopolistic competition markups,

$$1 - \tau_n^b = \frac{\epsilon_n^{L,b} - 1}{\epsilon_n^{L,b}} \frac{\sigma - 1}{\sigma} \ \forall n, b$$

$$\text{and, from equation (22)}, \ 1 + r_n^b = \frac{\sigma}{\sigma - 1} \mathcal{MC}^b \ \forall n, b.$$

Under this policy, all banks in a given city charge the same markup $\sigma/(\sigma - 1)$ over their respective marginal costs, eliminating the variation in markups that arises from differences in local market shares.

Equalizing markups reduces MPK dispersion by 1.5% and the average MPK by 1.7%, leading to an increase in GDP of 0.52%. When markups are equalized, capital flows to cities where markups were high (and investment was depressed) from cities where markups were low. Figure 6b shows that, as with interbank frictions, the effects are spatially heterogeneous. Market power hinders investment the most in the least productive cities, where banks tend to hold larger market shares. Capitalist welfare increases by 1% on average, while workers see a small after-tax welfare decline of 0.02%, despite a pre-tax welfare gain of 0.57%.

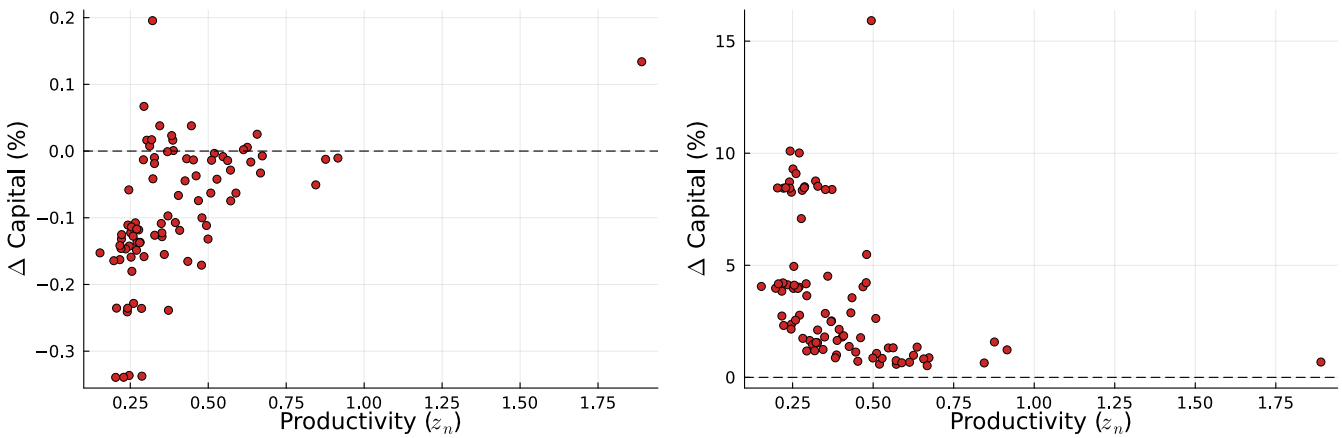
No markups. To provide an upper bound on the effect of market power, we eliminate markups entirely using subsidies that set interest rates equal to marginal cost,

$$1 - \tau_n^b = \frac{\epsilon_n^{L,b} - 1}{\epsilon_n^{L,b}} \ \forall n, b$$

$$\text{and, from equation (22)}, \ 1 + r_n^b = \mathcal{MC}^b \ \forall n, b.$$

Eliminating markups reduces MPK dispersion by 1.5% and the average MPK by 6%, as shown in Figure 4, leading to an increase in aggregate productivity of 3.55%. Workers' pre-tax welfare rises by 3.6%, but their

after-tax welfare falls by 0.02%. Average capitalist welfare rises by 4.1%.



(a) No financial frictions

(b) Constant markups

Figure 6: Heterogeneous Investment Responses

6.1 Financial integration, distance, and the propagation of productivity shocks

A key prediction of our model is that the branch network creates financial linkages between cities that operate independently of geographic proximity. To assess the quantitative importance of this channel relative to standard spatial linkages, which are mediated by distance, we shock the productivity z_n of each city n in turn by 100%, solve for the counterfactual equilibrium, and record the percentage change in employment and capital in every other city $j \neq n$. For each experiment, we construct two measures of bilateral financial integration between the shocked city n and each other city j,

$$\mathrm{FI}^D_{nj} = \frac{\sum_{b \in \mathcal{B}^n} \frac{\Delta D^b_n}{D^b_n} L^b_j}{\sum_b L^b_j}, \qquad \mathrm{FI}^L_{nj} = \frac{\sum_{b \in \mathcal{B}^n} \frac{\Delta L^b_n}{L^b_n} L^b_j}{\sum_b L^b_j}, \tag{36}$$

where $\Delta D^b_n / D^b_n$ and $\Delta L^b_n / L^b_n$ are the percentage changes in deposits and loans at bank b in the shocked city n, and L^b_j is bank b's baseline lending in city j. Both measures weight each bank's presence in city j by the intensity of the shock's impact on that bank in city n. The first measure, FI^D_{nj}, captures exposure through the deposit channel, while FI^L_{nj} captures exposure through competition for loans.

Pooling across all N experiments, we estimate a regression of the normalized employment elasticity $(\Delta \ell_j / \ell_j)/(\Delta z_n / z_n)$ on quintile dummies for bilateral travel time τ_{nj}, FI^D_{nj}, and FI^L_{nj}, controlling for shocked-city fixed effects. Quintiles are computed within each experiment, so that the variation in all three dimensions is not driven by differences across shocked cities.

Figure 7 plots the estimated coefficients for employment with 95% confidence intervals. The left panel shows that employment spillovers decay smoothly with travel time, consistent with standard spatial forces

operating through trade and migration: cities that are proximate to the shocked city benefit from the increase in productivity, as their market access has improved. The center panel shows that cities where banks receive very few of their deposits from the shocked area have lower employment responses. The result are smaller and concentrated in this quintile. The right panel shows a similar result. In cities where the present banks give few loans in the shocked city face lower competition for capital and have relatively higher employment responses.

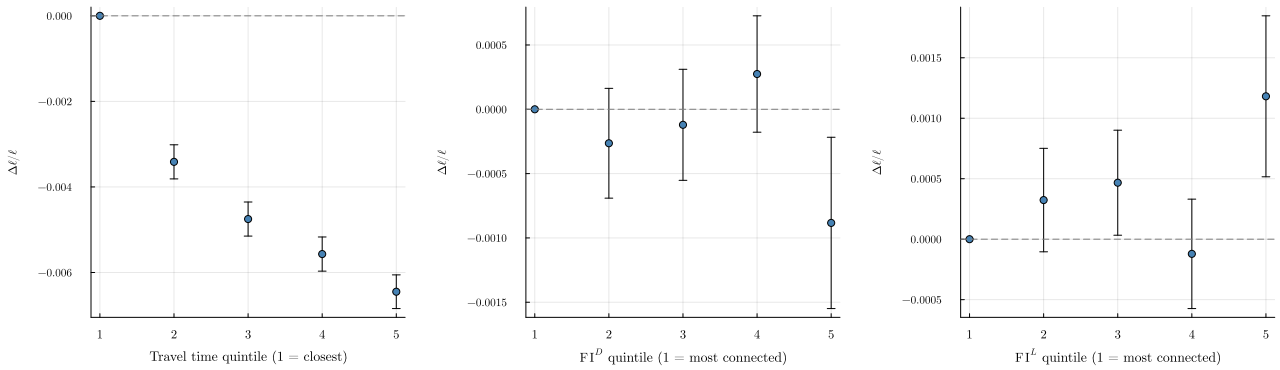


Figure 7: Propagation of productivity shocks: employment responses

7 Bank mergers

While our results in Section 6 suggest an important role for banks' market power, city-bank specific subsidies are rarely used in practice. On the other hand, evaluating and regulating bank mergers are recurrent questions facing policymakers. In Chile alone, four large bank mergers occurred during 2000-2020 (Marivil et al., 2021).

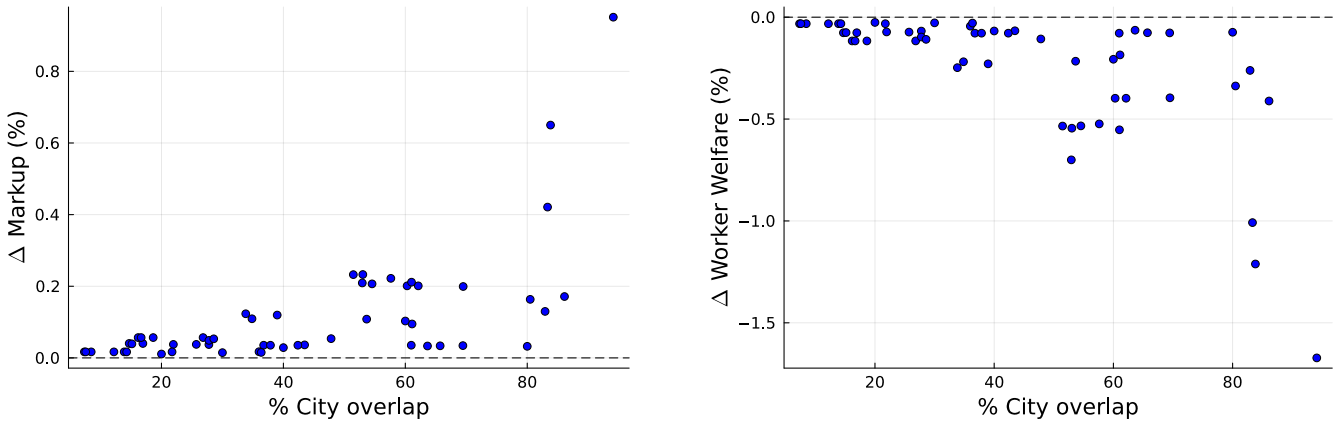
A natural concern with bank mergers is that lower competition will lead to higher interest rates. In Chile, where the median number of banks per city is three, bank mergers could lead to a substantial increase in markups in cities where both merging banks were present before the merger. Bank mergers, on the other hand, can enhance the efficiency of the banking sector if they allow the merging banks to circumvent the interbank market. Our framework with oligopolistic competition allows us to capture both sides of the trade-off.

Using the quantified version of our model, we evaluate every possible two-bank merger between the twelve largest banks in our data, leading to sixty-six mergers. For each merger, we compute the steady state of the economy where the two banks merge. We assume that the city-bank match, γ_n^b and κ_n^b of the merged bank equals the maximum among the two merging banks in city n whenever both banks are present in the city. We focus on worker welfare, productivity, and the average markup for each merger.

The economic effects of mergers are heterogeneous. The change in welfare ranges between -1.7% and $\approx 0\%$; the increase in markups varies between $\approx 0\%$ and 0.95%.

A first determinant of the welfare effect of a merger is the overlap in space of the two merging banks. We calculate geographic overlap as the percentage of cities where banks overlap relative to the largest number of cities in which each of the merging banks has branches.[20] If both banks have branches in the same cities, the merger will lead to a strong increase in markups without improving financial integration.

Figure 8 shows the effects of each merger as a function of city overlap. Figure 8a shows that the increase in markups is higher when banks' overlap in more cities. Figure 8b shows that the increase in markups is a key driver of welfare losses: higher markups lead to lower worker welfare.



(a) Δ Markup

(b) Δ Worker Welfare

Figure 8: Mergers' outcomes as a function of geographic overlap

We then sort mergers according to the financial integration dimension. If the two merging banks have opposite positions with the interbank market, merging allows them to transfer funds internally, bypassing the frictions associated with the interbank market. We define a measure of differences in merging banks' position in the interbank market as

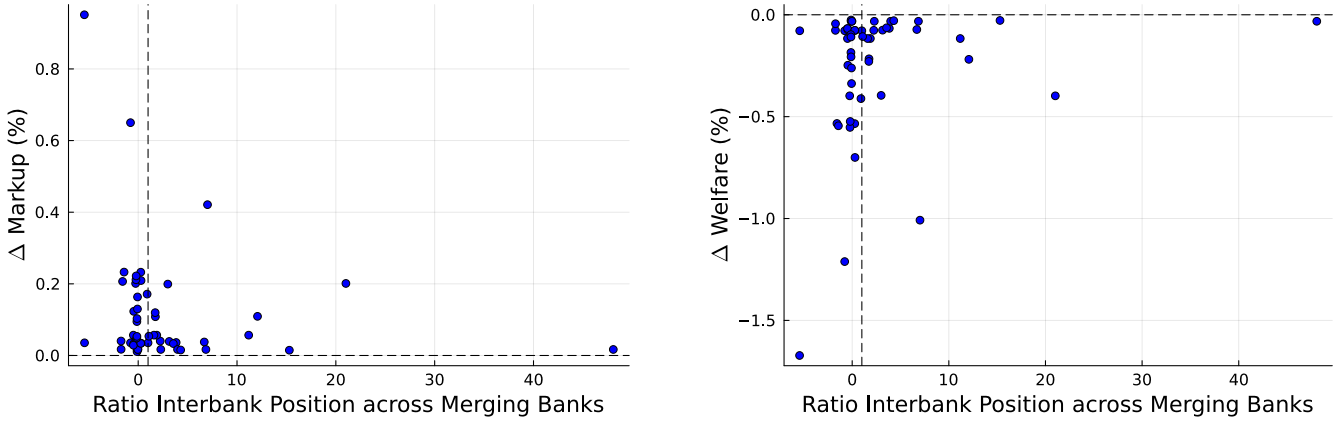
$$\text{Ratio of interbank market positions between A and B} = \frac{\min\left(F^A, F^B\right)}{\max\left(F^A, F^B\right)}.$$

The ratio takes the value of one if both banks have the same position in the interbank market; it is positive if both banks are borrowing or lending, and negative if they take opposite positions in the interbank market. The ratio is large in absolute value if the gap between banks' positions is large.

Whenever both banks' reliance on the interbank market is similar, merging will not change their reliance

[20] That is, if Bank A, present in 10 cities, merges with Bank B, present in 8 cities, and the overlap in 5 cities, the city overlap would be 50%.

on it. Indeed, Figure 9 shows that markup reductions and welfare effects are larger when merging banks have opposite positions in the interbank market. Merging allows these banks to circumvent the interbank market.



(a) Δ Markup

(b) Δ Worker Welfare

Figure 9: Mergers' outcomes as a function of banks' position in the interbank market

8 Conclusion

In this paper we make two main contributions. Empirically, we show that localized deposit shocks lead to more lending and reductions in interest rates by exposed banks. Moreover, we show that loan growth is concentrated in cities where banks hold a low share of the loan market. Our results complement and extend findings on banks' lending responses to deposit inflows. Previous studies did not study the interest rate response nor include banks' local market power into the analysis (Becker, 2007; Gilje et al., 2016; Bustos et al., 2020). Our results align with studies in industrial organization and finance which incorporate banks' local market power (Wang et al., 2020; Aguirregabiria et al., 2025).

We develop a novel quantitative spatial model with b banks, oligopolistic competition in credit markets and interbank frictions which rationalizes our empirical results and allows us to study their general equilibrium implications. We use the quantified version of our model to isolate two features of the observed branch network and their effect on economic outcomes in Chile. First, we study a counterfactual economy without interbank frictions and find that these frictions affect the allocation of capital across banks, which results in a negligible decline in aggregate productivity. Local market power reduces productivity by 0.5% by distorting the allocation of capital across cities. Finally, we use the model to assess all possible two-bank mergers in Chile. Bank mergers are a common challenge for policy makers: in Chile alone, four large banks merged between 2000 and 2020 (Marivil et al., 2021). A spatial perspective sheds light on the trade-offs

associated with bank mergers. Our main results are that the welfare effects of bank mergers are higher when the merging banks have little geographic overlap, which limits the increase in markups, and when one of them borrows and the other lends in the interbank market. In such cases, merging allows them to channel funds internally and bypass the frictions associated with the interbank market.

An important limitation of our analysis is that we take the bank branch network as given throughout. This means that our counterfactual exercises capture the effects of interbank frictions and market power conditional on the observed geographic distribution of branches, but cannot account for how banks would adjust their entry and exit decisions in response to changes in the competitive environment. This limitation is particularly relevant for the policy implications of our market power counterfactuals. For example, under constant markups (Figure 6b), most of the increase in capital is directed toward lower-productivity cities. In practice, however, policies that tax oligopolistic rents may also reduce banks' incentives to maintain branches in these cities, potentially undoing some of the gains from lower markups. A full analysis of competition policy in banking should therefore incorporate banks' endogenous branching decisions, as in Oberfield et al. (2024). Integrating entry and exit with oligopolistic pricing in a spatial framework is a challenging but important direction for future research.

References

Acemoglu, Daron and Melissa Dell, "Productivity Differences between and within Countries," *American Economic Journal: Macroeconomics*, January 2010, *2* (1), 169–88.

Aguirregabiria, Victor, Robert Clark, and Hui Wang, "The Geographic Flow of Bank Funding and Access to Credit: Branch Networks, Synergies, and Local Competition," *American Economic Review*, 2025, *forthcoming.*

Albertazzi, Ugo, Finn Faber, Alessandro Gavazza, Oana Maria Georgescu, and Ernest Lecomte, "Bank Deposit Pricing in the Euro Area," Working Paper 2024.

Allen, Treb and Costas Arkolakis, "Quantitative Regional Economics," Working Paper 33436, National Bureau of Economic Research January 2025.

Altavilla, Carlo, Miguel Boucinha, and Paul Bouscasse, "Supply or Demand: What Drives Fluctuations in the Bank Loan Market?," ECB Working Paper Series 2646, European Central Bank February 2022. ECB Working Paper Series, No. 2646.

Andersen, Leif, Darrell Duffie, and Yang Song, "Funding Value Adjustments," *The Journal of Finance*, 2019, *74* (1), 145–192.

Ashcraft, Adam B., "Are Banks Really Special? New Evidence from the FDIC-Induced Failure of Healthy Banks," *American Economic Review*, December 2005, *95* (5), 1712–1730.

Atkeson, Andrew and Ariel Burstein, "Pricing-to-Market, Trade Costs, and International Relative Prices," *American Economic Review*, December 2008, *98* (5), 1998–2031.

Baqaee, David R. and Ariel Burstein, "Aggregate Productivity with Discrete Choice," March 2026. UCLA Working Paper.

Becker, Bo, "Geographical segmentation of US capital markets," *Journal of Financial Economics*, 2007, *85* (1), 151–178.

Bergquist, Lauren Falcao, Danial Lashkari, and Eric Verhoogen, "Wedges: A Microeconomic Perspective on Misallocation," in Pascaline Dupas, Penny Goldberg, and Rohini Pande, eds., *Handbook of Development Economics*, 2026. Forthcoming in Handbook of Development Economics, Volume 6. Dated Jan. 29, 2026 (version in linked PDF).

Bordeu, Olivia, Gustavo González, and Marcos Sorá, "Bank Competition and Investment Costs across Space," Technical Report, Banco Central de Chile April 2025.

Bronnenberg, Bart J., Sanjay K. Dhar, and Jean-Pierre Dubé, "Consumer Packaged Goods in the United States: National Brands, Local Branding," 2007.

Bustos, Paula, Gabriel Garber, and Jacopo Ponticelli, "Capital Accumulation and Structural Transformation*," *The Quarterly Journal of Economics*, 01 2020, *135* (2), 1037–1094.

Caliendo, Lorenzo, Fernando Parro, Esteban Rossi-Hansberg, and Pierre-Daniel Sarte, "The Impact of Regional and Sectoral Productivity Changes on the U.S. Economy," *The Review of Economic Studies*, 12 2017, *85* (4), 2042–2096.

Conley, Timothy G. and Giorgio Topa, "Socio-economic distance and spatial patterns in unemployment," *Journal of Applied Econometrics*, 2002, *17* (4), 303–327.

Corbae, Dean and Pablo D'Erasmo, "A Quantitative Model of Banking Industry Dynamics," *Journal of Political Economy Macroeconomics*, 2025. Forthcoming.

Crawford, Gregory S., Nicola Pavanini, and Fabiano Schivardi, "Asymmetric Information and Imperfect Competition in Lending Markets," *American Economic Review*, July 2018, *108* (7), 1659–1701.

D'Amico, Leonardo and Maxim Alekseev, "Capital Market Integration and Growth Across the United States," Working Paper November 2024.

Degryse, Hans and Steven Ongena, "Distance, Lending Relationships, and Competition," *The Journal of Finance*, 2005, *60* (1), 231–266.

Desmet, Klaus and Esteban Rossi-Hansberg, "Urban Accounting and Welfare," *American Economic Review*, October 2013, *103* (6), 2296–2327.

Drechsler, Itamar, Alexi Savov, and Philipp Schnabl, "The Deposits Channel of Monetary Policy," *The Quarterly Journal of Economics*, 2017, *132* (4), 1819–1876.

Garmaise, Mark J. and Tobias J. Moskowitz, "Bank Mergers and Crime: The Real and Social Effects of Credit Market Competition," *The Journal of Finance*, 2006, *61* (2), 495–538.

Gilje, Erik P., Elena Loutskina, and Philip E. Strahan, "Exporting Liquidity: Branch Banking and Financial Integration," *The Journal of Finance*, 2016, *71* (3), 1159–1184.

Hanson, Samuel G., Andrei Shleifer, Jeremy C. Stein, and Robert W. Vishny, "Banks as patient fixed-income investors," *Journal of Financial Economics*, 2015, *117* (3), 449–469.

Henderson, J. Vernon, "The Sizes and Types of Cities," *American Economic Review*, 1974, *64* (4), 640–56.

Herreno, Juan, "Aggregating the Effect of Bank Credit Supply Shocks on Firms," October 2023. Working paper, UC San Diego. October 5, 2023.

Hsieh, Chang-Tai and Peter J. Klenow, "Misallocation and Manufacturing TFP in China and India," *The Quarterly Journal of Economics*, 2009, *124* (4), 1403–1448.

Kashyap, Anil K., Raghuram Rajan, and Jeremy C. Stein, "Banks as Liquidity Providers: An Explanation for the Coexistence of Lending and Deposit-taking," *The Journal of Finance*, 2002, *57* (1), 33–73.

Kleinman, Benny, Ernest Liu, and Stephen J. Redding, "Dynamic Spatial General Equilibrium," *Econometrica*, 2023, *91* (2), 385–424.

Lucas, Robert E., "Why Doesn't Capital Flow from Rich to Poor Countries?," *The American Economic Review*, 1990, *80* (2), 92–96.

Maingi, Quinn, "Regional Banks, Aggregate Effects," *Journal of Financial Economics*, 2026, *176*, 104226.

Marivil, Gonzalo, José Matus, and Daniel Oda, "Caracterización del sector bancario chileno: 1990-2020," December 2021. Mimeo in *Intermediación Financiera y Banca Central en Chile*.

Monte, Ferdinando, Stephen J. Redding, and Esteban Rossi-Hansberg, "Commuting, Migration, and Local Employment Elasticities," *American Economic Review*, December 2018, *108* (12), 3855–90.

Morelli, Juan, Matias Moretti, and Venky Venkateswaran, "Geographical Diversification in Banking: A Structural Evaluation," Working Paper March 2024.

Nguyen, Hoai-Luu Q., "Are Credit Markets Still Local? Evidence from Bank Branch Closings," *American Economic Journal: Applied Economics*, January 2019, *11* (1), 1–32.

Oberfield, Ezra, Esteban Rossi-Hansberg, Nicholas Trachter, and Derek Wenning, "Banks in Space," Technical Report, Working Paper 2024.

Pellegrino, Bruno, Enrico Spolaore, and Romain Wacziarg, "Barriers to Global Capital Allocation*," *The Quarterly Journal of Economics*, 06 2025, *140* (4), 3067–3131.

Petersen, Mitchell A. and Raghuram G. Rajan, "Does Distance Still Matter? The Information Revolution in Small Business Lending," *The Journal of Finance*, 2002, *57* (6), 2533–2570.

Quincy, Sarah and Chenzi Xu, "Branching Out: Capital Mobility and Long-Run Growth," Technical Report Working Paper 34457, National Bureau of Economic Research (NBER) November 2025.

Redding, Stephen J. and Esteban Rossi-Hansberg, "Quantitative Spatial Economics," *Annual Review of Economics*, 2017, *9* (1), 21–58.

Scharfstein, David and Adi Sunderam, "Market Power in Mortgage Lending and the Transmission of Monetary Policy," Technical Report, Working Paper 2016.

Wang, Yifei, Toni M. Whited, Yufeng Wu, and Kairong Xiao, "Bank Market Power and Monetary Policy Transmission: Evidence from a Structural Estimation," Technical Report, Working Paper 2020.

9 Appendix

A Empirical appendix

A.1 Chile's financial development

We use public data from the World Bank, accessed online on June 2024. Figure 10 below shows the evolution of the two indicators of financial development mentioned in the main text.

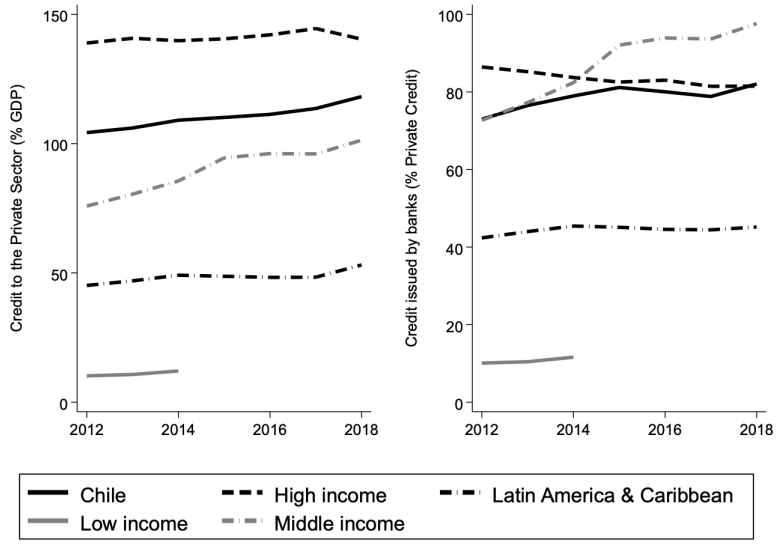


Figure 10: Financial development

A.2 The importance of banks for domestic credit in Chile: Survey evidence

Firms and households rely mostly on banks for financial services and local branches play a significant role.

Firms. To delve deeper into the importance of banks for private firms in Chile, we rely on firm-level data from the 2015 *Encuesta Longitudinal de Empresas* (ELE), a nationally representative survey that includes a module on firms' sources of credit. We calculate the percentage of private firms that borrow from banks and the percentage of firms for which banks constitute the main source of credit. We exclude Santiago, the capital city and home to approximately 29% of the population and bigger firms, to show that Santiago does not drive the results. The first two columns of Table 5 show that banks stand out as the main source of credit for large private firms outside the capital area.

Table 5: Credit sources for firms (excluding Santiago)

Firm size	2015 ELE		
	% borrows from banks	% biggest loan comes from banks	% private employment
Micro	57.1	16.7	7.7
Small	66.4	29.6	39.3
Medium	77.7	42.1	21.9
Large	80.5	50.4	30.1

Households. In 2007 and 2017, the *Encuesta Financiera de Hogares* (EFH), a nationally representative survey of households' financial behavior, included modules on the financial assets held by households; using

these modules, we first document that households rely significantly on banks to purchase financial assets (compared to other institutions) and, secondly, that Internet banking remains limited.

In the EFH we separately observe the total amount invested by an individual household in stocks, mutual funds, fixed income, saving accounts, and other instruments. The survey contains information on the financial institution through which these assets were purchased. Panel A in Table 6 shows — for the sub-sample of respondents with positive financial assets — what percentage of savings were allocated to each asset and the percentage of respondents who used banks to purchase that asset. Banks are the primary institutions used by households to invest in mutual funds and fixed-income assets and to open savings accounts. These represent around half the total investment in financial assets in 2007 and 2017.

The main concern regarding reliance on local branches is the expansion of Internet banking, which makes it easier to save and borrow from geographically distant banks. The EFH includes a question on the use of Internet banking, where people are asked whether they used the Internet to carry out a variety of financial transactions. Panel B in Table 6 shows the share of respondents who used the Internet to purchase financial assets or get new loans. In both cases, we calculate the percentage over the total number of respondents who either purchase assets or get new loans. Internet was used more intensively to purchase new financial assets than to get loans. Although there was an increase in both uses between 2007 and 2017, a majority of the transactions still happen in physical branches. Moreover, the survey does not distinguish between new transactions and the first transaction with a bank, therefore representing an upper bound on the reliance on the Internet to start new financial relationships with an institution.

Table 6: Households' savings behavior

A. Asset types	2007 EFH		2017 EFH	
	% of assets	% purchased through banks	% of assets	% purchased through banks
Stock	19.1	36.1	15.1	44.2
Mutual Fund	30.8	80.4	24.3	83.7
Fixed-income	9.4	82.9	21.3	90.0
Saving Account	7.0	91.6	7.3	72.3
Other	33.6	-	31.7	-
B. Used the internet to...	% respondents in 2007		% respondents in 2017	
purchase financial assets	6.5		21.0	
get a loan	0.3		2.1	

A.3 Concentration in banking industry

We calculate the market share for top banks using aggregate data from the CMF. Results are shown in Figure 11.

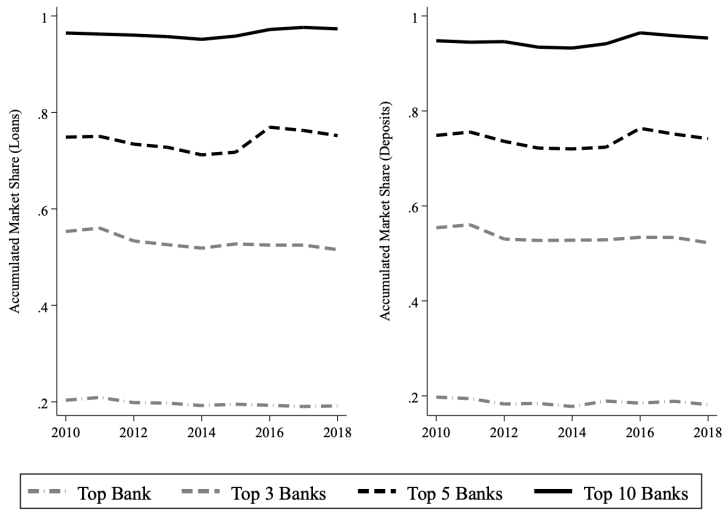


Figure 11: Concentration in the Banking Industry

A.4 Spatial Clustering of Banks

To determine whether banks' economic activity is geographically clustered we follow the approach in Conley and Topa (2002), who study the degree of spatial correlation in unemployment between neighborhoods. More closely related to our setting, the approach has been used to study the degree of geographical concentration in market shares for a variety of consumer goods in Bronnenberg et al. (2007). For this exercise, we use aggregate data from the year 2015 (publicly available through the CMF) and focus exclusively on banks present in at least ten cities in 2015. These banks explained 96.8% of all the outstanding loans in that year. We exclude the metropolitan area around Santiago.

Extensive margin. First, we define the dummy variable X_{ib}, which takes the value 1 if bank b gave any loans in city i. We are interested in the correlation of X_{ib} between pairs of cities i, j as the distance between i and j changes. Figure 12 shows these correlations for each individual bank, where we have defined bins of 250 kilometers in size.

A correlation close to zero suggests that banks' presence is independent across cities. To determine how close to zero the observed measures of correlation would be if the X_{ib} were independent we follow the bootstrap approach in Conley and Topa (2002). We create 100 samples in which we randomize the identity of the cities in which each bank is present by drawing (with replacement) from the observed distribution of that particular bank. The two dashed lines in each figure show the 90% confidence interval across bootstrapped samples. For almost all banks and all distance bins we cannot reject that the observed correlations are different than what we would observe if banks' presence was independent across cities.

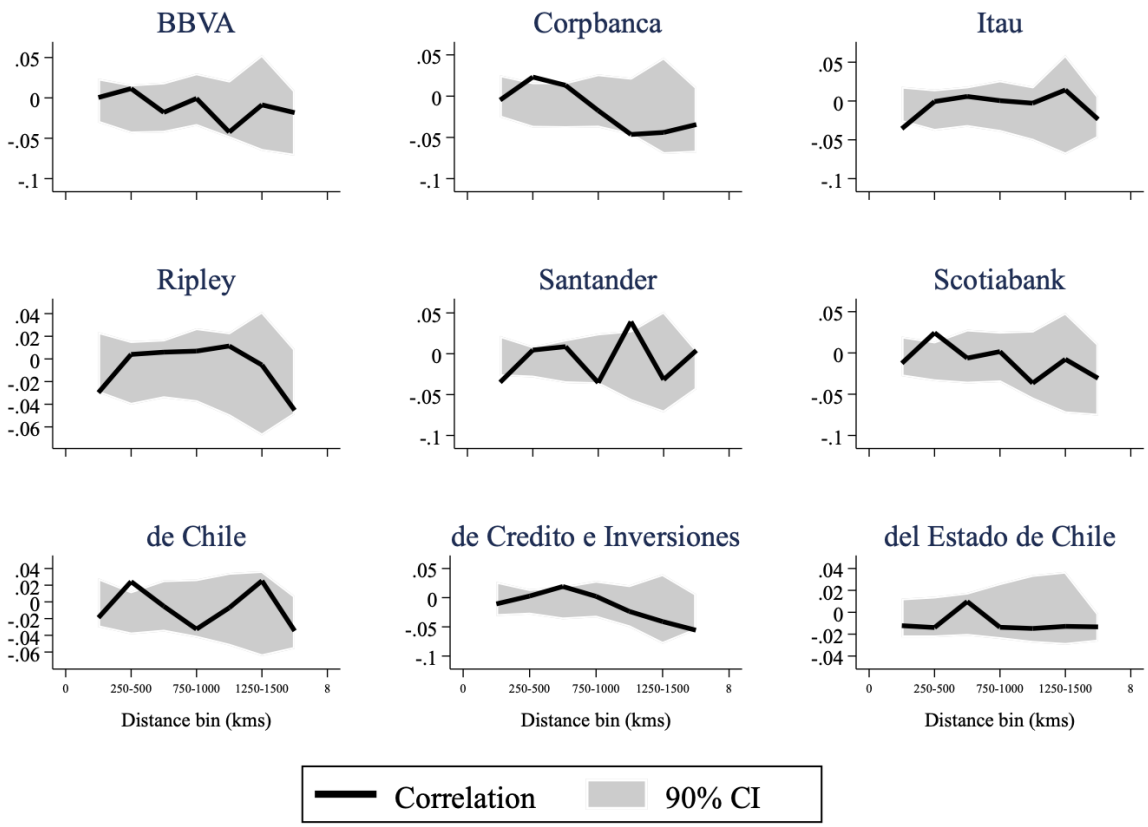


Figure 12: Spatial Correlation in Bank's Presence (Extensive Margin)

Intensive margin. To complement the previous analysis, we study whether there is spatial correlation in market shares (conditional on banks' presence). The approach is analogous to the one described above except that, in this case, the outcome variable is defined as the share of outstanding loans in city i issued by bank b in 2015. When we construct the confidence intervals, we randomize the particular market share of a bank in a city without changing the cities in which a bank is present, therefore focusing exclusively on the intensive margin.

Figure 13 shows the results. The conclusion is similar to the one before, albeit less clear-cut. *Banco de Crédito e Inversiones* and *Banco Santander* exhibit patterns of geographical clustering in market shares.

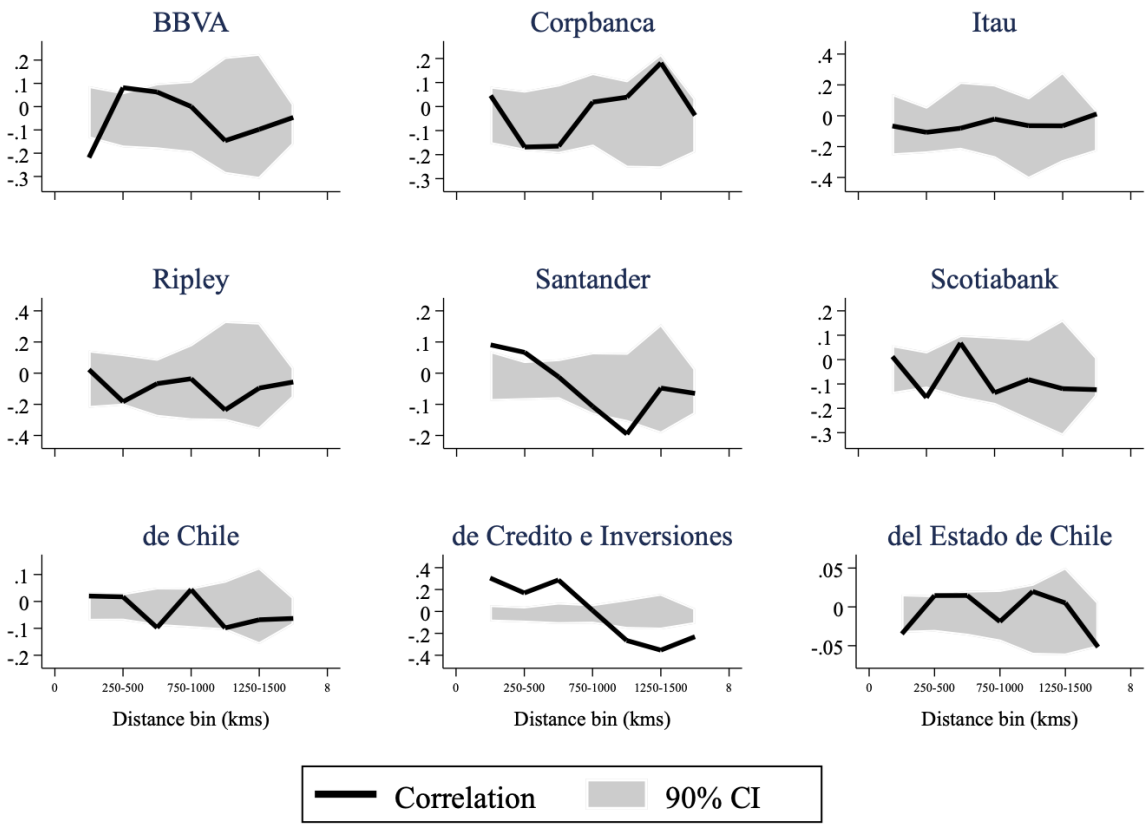


Figure 13: Spatial Correlation in Loan Market Shares (Intensive Margin)

A.5 Details on the loan subsample for interest rate analysis

The administrative dataset includes, for each loan, the size and sector of the borrowing firm, the type of loan, its maturity, and amount. Moreover, the dataset includes two measures of risk at the loan level. The first is a categorical risk rating assigned by banks when a firm applies for a loan. For large firms, banks conduct individual assessments, classifying them into one of 16 risk categories: A1–A6 (low risk), B1–B4, and C1–C6 (high risk). For smaller firms, the risk is assessed after the banks classify firms with similar characteristics together. The second measure of risk is the expected loss on each loan, reflecting the bank's projection of potential default costs.[21]

We restrict the sample to loans denominated in Chilean pesos and not associated with any public guarantee. We keep fixed-interest rate loans and exclude loans issued by *BancoEstado* and drop the first loan in a firm-bank relationship, where the interest rate may be systematically different given that the bank lacks information about the firm. By omitting these loans, our interest rates results should be interpreted as declines in interest rates that banks offer to firms with extensting relationships.

[21]This study was developed within the scope of the research agenda conducted by the Central Bank of Chile (CBC) in economic and financial affairs of its competence. The CBC has access to anonymized information from various public and private entities, by virtue of collaboration agreements signed with these institutions. To secure the privacy of workers and firms, the CBC mandates that the development, extraction and publication of the results should not allow the identification, directly or indirectly, of natural or legal persons. Officials of the Central Bank of Chile processed the disaggregated data. All the analysis was implemented by the authors and did not involve nor compromise the SII, the CMF, and AFC. The information contained in the databases of the Chilean IRS is of a tax nature originating in self-declarations of taxpayers presented to the Service; therefore, the veracity of the data is not the responsibility of the Service.

A.6 Details on the Shift-Share IV

We use data from the IMF Commodity Price series. Figure 14 shows the evolution of the world price of salmon at a monthly frequency.

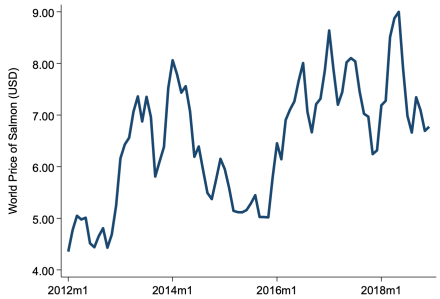


Figure 14: World Price of Salmon

Figure 15 shows the share of local employment in the Fishing industry. The industry is concentrated in the Southern region.

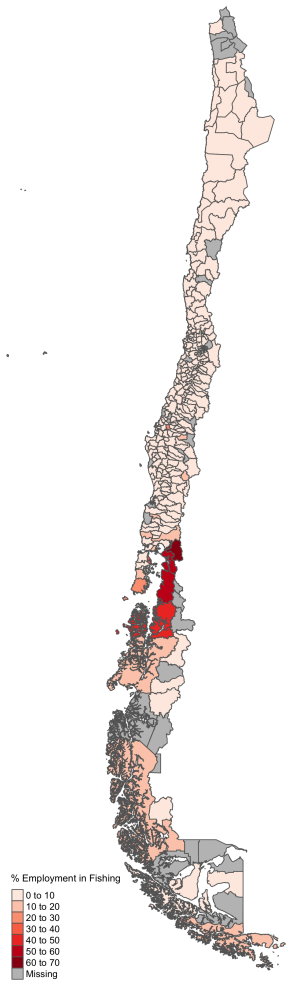


Figure 15: Share of local employment in the fishing industry

A.7 Robustness and extensions of the empirical analysis in Section 3

Robustness 1: Controlling for city-time unobservable shocks. We first modify our main estimating equation, equation (1), by dropping city-month fixed effects but including city fixed effects which absorb time-invariant differences between cities. Table 7 below shows that our estimate of the elasticity increases from 0.11 to 0.13. This difference suggest when exposed banks increase lending in destination cities, all banks present in that city respond by lending more. In our main specification, city-month fixed effects partly absorb those aggregate effects.

Table 7: The Effect of Deposit Shocks on Lending

	Loans (log)			
	(1)	(2)	(3)	(4)
Deposits (log)	0.133***	0.147***	0.123***	0.122***
	(0.048)	(0.048)	(0.040)	(0.040)
× Distance		−0.010		
		(0.011)		
× Mkt Share			−0.439***	−0.427***
			(0.122)	(0.124)
× D-L Ratio				−0.000
				(0.000)
× Mkt Share × D-L Ratio				−0.000
				(0.000)
Loans_{t-4} (log)	0.672***	0.671***	0.586***	0.586***
	(0.032)	(0.032)	(0.037)	(0.037)
City × Bank FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes
Observations	30,353	30,353	30,021	30,021
R-squared	0.994	0.994	0.995	0.995

Notes: This table reports 2SLS estimates of the effect of deposit shocks on bank lending. The sample excludes cities with any employment in fishing industries. Distance is average travel time to fishing cities. Market share is bank b's share of local lending in city n. D-L Ratio is the local deposit-to-loan ratio. Observations weighted by total bank-level loan volume. Standard errors clustered at the bank-month level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Robustness 2: Two-way clustering at the bank-month and city-bank level. In our baseline specification we cluster at the bank-month level, which is the level at which the deposit shocks occur. To account for correlation in shocks within city-bank pair across time, we cluster standard error at both levels. Table 8 shows the results. The main difference is that the interaction between deposit shocks and market share is no longer statistically significant.

Robustness 3: Loan flows as an outcome. The public data from the CMF reports the stock of outstanding loans and the value of loans maturing every month. In principle, therefore, we could calculate the flow of new loans by differencing the stock and adding the value of maturing loans. However, mainly due to noise in the data, this measure leads to many negative values. To address this issue, we add over a longer horizon.

For every t, we calculate an estimate of the number of new loans over the last three months as

$$\text{Loan Flow}_{ntb} = \text{Loan Stock}_{ntb} - \text{Loan Stock}_{nt-4b} + \sum_{j=1}^{3} \text{Maturing Loans}_{nt-sb}. \tag{37}$$

Even after averaging over 3 months, around 5% of the observations are negative. Table 9 shows the

Table 8: Robustness: Two-Way Clustering

	Loans (log)			
	(1)	(2)	(3)	(4)
Deposits (log)	0.111*	0.092	0.172***	0.162***
	(0.062)	(0.076)	(0.059)	(0.059)
× Distance		0.014		
		(0.030)		
× Mkt Share			−1.518***	−1.457***
			(0.328)	(0.316)
× D-L Ratio				0.004*
				(0.002)
× Mkt Share × D-L Ratio				−0.001*
				(0.001)
Loans_{t-4} (log)	0.684***	0.684***	0.517***	0.517***
	(0.049)	(0.049)	(0.050)	(0.050)
City × Bank FE	Yes	Yes	Yes	Yes
City × Month FE	Yes	Yes	Yes	Yes
Observations	28,439	28,439	28,127	28,127
R-squared	0.994	0.994	0.996	0.996

Notes: This table replicates Table 2 with two-way clustering at the bank-month and city-bank levels. Observations weighted by total bank-level loan volume. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

results of estimating equation (1) with flows as an outcome variable. The results are qualitatively similar to those in the baseline analysis but larger quantitatively.

Table 9: Robustness: Loan Flows as Outcome

	Loan Flows (log)			
	(1)	(2)	(3)	(4)
Deposits (log)	0.788***	0.480***	0.996***	0.922***
	(0.133)	(0.167)	(0.117)	(0.117)
× Distance		0.231***		
		(0.050)		
× Mkt Share			−3.635***	−3.468***
			(0.546)	(0.550)
× D-L Ratio				0.046***
				(0.009)
× Mkt Share × D-L Ratio				−0.007***
				(0.001)
Loans_{t-4} (log)	0.363***	0.364***	−0.007	0.001
	(0.042)	(0.043)	(0.053)	(0.052)
City × Bank FE	Yes	Yes	Yes	Yes
City × Month FE	Yes	Yes	Yes	Yes
Observations	26,556	26,556	26,253	26,253
R-squared	0.909	0.909	0.914	0.915

Notes: This table replicates Table 2 using log loan flows rather than log loan stocks as the outcome variable. Observations weighted by total bank-level loan volume. Standard errors clustered at the bank-month level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B Mathematical appendix

B.1 Workers

Starting from equation (6) in the main text we derive steady-state employment shares. Using properties of the T1EV distribution of idiosyncratic shocks and dropping time-subindices (as we focus on a steady state), the value function of a worker who has moved to n is

$$v_n = \ln b_{nt} + \ln \frac{w_n(1-\tau^{ss})}{P_{nt}} + \rho \ln \Big(\sum_{d=1}^{N} exp\Big(\frac{\beta}{\rho} v_d\Big)\Big).$$

Then,

$$exp\Big(\frac{\beta}{\rho} v_n\Big) = b_n^{\frac{\beta}{\rho}} \times [w_n(1-\tau^{ss})]^{\frac{\beta}{\rho}} \times P_n^{\frac{-\beta}{\rho}} \times \Big(\sum_{d=1}^{N} exp\Big(\frac{\beta}{\rho} v_d\Big)\Big)^{\beta}.$$

We define

$$\phi \equiv \sum_{d=1}^{N} exp\Big(\frac{\beta}{\rho} v_d\Big). \tag{38}$$

The steady-state value of ϕ solves

$$\phi = \sum_{d=1}^{N} b_d^{\frac{\beta}{\rho}} \times [w_d(1-\tau^{ss})]^{\frac{\beta}{\rho}} \times P_d^{\frac{-\beta}{\rho}} \times \phi^{\beta} \tag{39}$$

$$= \Big(\sum_{d=1}^{N} b_d^{\frac{\beta}{\rho}} \times [w_d(1-\tau^{ss})]^{\frac{\beta}{\rho}} \times P_d^{\frac{-\beta}{\rho}} \Big)^{\frac{1}{1-\beta}} \tag{40}$$

From the T1EV assumption for idiosyncratic shocks, migration shares between any cities n and d are

$$M_{nd} = \ell_d = \frac{exp\Big(\frac{\beta}{\rho} v_d\Big)}{\sum_{m=1}^{N} exp\Big(\frac{\beta}{\rho} v_m\Big)} = exp\Big(\frac{\beta}{\rho} v_d\Big)\phi^{-1} = b_d^{\frac{\beta}{\rho}} \, [w_d(1-\tau^{ss})]^{\frac{\beta}{\rho}} \, P_d^{\frac{-\beta}{\rho}} \, \phi^{\beta-1}.$$

Given that we have normalized the population to 1, migration shares and population equalize in the steady state. The expression for population in the main text, equation (29), follows.

B.2 Capitalists

For this subsection we drop n from the sub-indices for clarity, as the problem is isomorphic for all capitalists. The problem of the capitalist can be divided in two stages. In a first stage, the capitalist decides from which banks to borrow in order to finance a level of investment i_t at the lowest cost. In a second stage she maximizes her welfare by deciding how much investment to make taking the cost of investment, $\mathcal{L}_t(i_t)$, as given. We begin by solving the latter.

Solving for $\mathcal{L}_t(i_t)$ The problem of minimizing the cost of investment is

$$\mathcal{L}_t(i_t) = \min_{\{L_{t+1}^b\}_b} \sum_{b \in \mathcal{B}} L_{t+1}^b (1 + r_{t+1}^b) \tag{41}$$

$$s.t. \;\; i_t = \left[\sum_{b \in \mathcal{B}} (\gamma^b \tfrac{L_{t+1}^b}{P_t})^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \tag{42}$$

From the first order condition with respect to an arbitrary L_{t+1}^b,

$$\mu (\tfrac{\gamma^b}{P_t})^{\frac{\sigma-1}{\sigma}} \Big(\frac{i_t}{L_{t+1}^b} \Big)^{\frac{1}{\sigma}} = (1 + r_{t+1}^b), \tag{43}$$

where μ is the multiplier associated with the constraint in equation (42). Taking the ratio of equation (43) for two banks b and b',

$$\frac{L_{t+1}^{b'}}{L_{t+1}^b} = \Big(\frac{1 + r_{t+1}^b}{1 + r_{t-1}^{b'}} \Big)^{\sigma} \Big(\frac{\gamma^{b'}}{\gamma^b} \Big)^{\sigma-1}. \tag{44}$$

From here, picking an arbitrary b':

$$i_t = \Big(\sum_{b \in \mathcal{B}} (\gamma^b \tfrac{L_{t+1}^b}{P_t})^{\frac{\sigma-1}{\sigma}} \Big)^{\frac{\sigma}{\sigma-1}} = (1 + r_{t+1}^{b'})^{\sigma} (\gamma^{b'})^{1-\sigma} \tfrac{L_{t+1}^{b'}}{P_t} \Big[\sum_{b \in \mathcal{B}} (\tfrac{1+r_{t+1}^b}{\gamma^b})^{1-\sigma} \Big]^{-\frac{\sigma}{1-\sigma}}. \tag{45}$$

Defining $R_{t+1} \equiv \Big[\sum_{b \in \mathcal{B}} (\frac{1+r_{t+1}^b}{\gamma^b})^{1-\sigma} \Big]^{\frac{1}{1-\sigma}}$,

$$i_t R_{t+1}^{\sigma} = (1 + r_{t+1}^b)^{\sigma} (\gamma^b)^{1-\sigma} \tfrac{L_{t+1}^b}{P_t} \tag{46}$$

and, therefore, we can express the equilibrium loans from bank b as

$$\frac{L^b_{t+1}}{P_t} = \Big(\frac{R_{t+1}}{1+r^b_{t+1}}\Big)^{\sigma} i_t(\gamma^b)^{\sigma-1}, \tag{47}$$

which shows as equation (**??**) in the main text. From equation (47) and the definition of $\mathcal{L}_t(i_t)$,

$$\mathcal{L}_t(i_t) = \sum_{b\in\mathcal{B}} L^b_{t+1}(1+r^b_{t+1}) = i_t R_{t+1} P_t. \tag{48}$$

which shows as equation (**??**) in the main text.

Capitalist's full problem The full problem of the capitalist is

$$\max_{\{C^c_t, D^b_{t+1}, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \Big[\log C^c_t + \alpha \log D_{nt+1}\Big]$$

$$\underline{s.t:}\ C^c_t + \sum_b \frac{D^b_{t+1}}{P_t} + \frac{(k_t - k_{t-1}(1-\delta))R_t P_{t-1}}{P_t} = \frac{\hat{r}_t k_t}{P_t} + \sum_b \frac{D^b_t}{P_t}(1+\tilde{r}^b_t) + \frac{T^c_t}{P_t} \tag{49}$$

$$D_{t+1} = \Big[\sum_b (\kappa^b D^b_{t+1})^{1-\frac{1}{\eta}}\Big]^{\frac{\eta}{\eta-1}} \tag{50}$$

$$k_0, \{D^b_0, L^b_0\}_b$$

where we have replaced $i_{t-1} = k_t - k_{t-1}(1-\delta)$ and expressed the budget constraint in real terms. The first-order conditions with respect to k_t, C^c_t and D^b_{t+1} are

$$\lambda_t \frac{\hat{r}_t}{P_t} + \lambda_{t+1}\frac{(1-\delta)R_{t+1}P_t}{P_{t+1}} = \lambda_t \frac{R_t P_{t-1}}{P_t}, \tag{51}$$

$$\frac{\beta^t}{C^c_t} = \lambda_t, \tag{52}$$

$$\text{and}\ \ \beta^t \alpha D_{t+1}^{\frac{1-\eta}{\eta}} (D^b_{t+1})^{-\frac{1}{\eta}} (\kappa^b)^{\frac{\eta-1}{\eta}} + \lambda_{t+1}\frac{1+\tilde{r}^b_{t+1}}{P_{t+1}} = \frac{\lambda_t}{P_t}. \tag{53}$$

Equation (51) equates the marginal benefit of an extra unit of capital in period t, which consists of the per-period rental rate and the extra capital she would carry to period $t+1$, to its cost, loan repayment in period t. The first order condition with respect to consumption, equation (52), is standard. The first order condition with respect to deposits in a specific bank, equation (53), reflects the dual role of deposits in the model: they enter directly into the utility function and are means for transferring resources between periods.

By combining equation (51) and equation (52) we derive the following Euler equation,

$$\frac{P_{t+1}C_{t+1}}{P_t C_t} = \beta(1-\delta)\frac{R_{t+1}P_t}{R_t P_{t-1} - \hat{r}_t}. \tag{54}$$

Replacing equation (52) into equation (53), and then replacing $C_{t+1}P_{t+1}$ from equation (54),

$$\frac{\alpha}{D_{t+1}}(\kappa^b)^{\frac{\eta-1}{\eta}}\left(\frac{D_{t+1}}{D^b_{t+1}}\right)^{\frac{1}{\eta}} = \frac{1}{P_t C_t}\left[1 - \frac{(1+\tilde{r}^b_{t+1})(R_t P_{t-1} - \hat{r}_t)}{(1-\delta)R_{t+1}P_t}\right]. \tag{55}$$

Dividing this equation for two banks, b and b',

$$\frac{D^b_{t+1}}{D^{b'}_{t+1}} = \left(\frac{\kappa^b}{\kappa^{b'}}\right)^{\eta-1} \left(\frac{q^b_{t+1}}{q^{b'}_{t+1}}\right)^{-\eta}, \tag{56}$$

where we defined q^b_{t+1} as

$$q^b_{t+1} \equiv 1 - \left(1 + \tilde{r}^b_{t+1}\right) / \left(\frac{(1-\delta)R_{t+1}P_t}{R_t P_{t-1} - \hat{r}_t}\right). \tag{57}$$

We define the deposit price index as

$$Q_{t+1} \equiv \left(\sum_b (\tfrac{q^b_{t+1}}{\kappa^b})^{1-\eta}\right)^{\frac{1}{1-\eta}}. \tag{58}$$

It follows from equation (56) and the definition of D_{t+1} that the supply of deposits to bank b is given by

$$D^b_{t+1} = D_{t+1}(\kappa^b)^{\eta-1} \left(\frac{Q_{t+1}}{q^b_{t+1}}\right)^{\eta}. \tag{59}$$

Replacing this back into equation (55) we obtain that the capitalist equalizes of expenditure on the two 'goods' available to her, consumption and deposits,

$$D_{t+1}Q_{t+1} = \alpha P_t C_t. \tag{60}$$

The nominal value of total deposits at t is given by

$$\sum_b D^b_{t+1} = \sum_b D_{t+1}(\kappa^b)^{\eta-1}(\tfrac{Q_{t+1}}{q^b_{t+1}})^{\eta} = D_{t+1}Q^{\eta}_{t+1} \overbrace{\sum_b (\kappa^b)^{\eta-1}(q^b_{t+1})^{-\eta}}^{\equiv \tilde{Q}_{t+1}}, \tag{61}$$

where $\tilde{Q}_{t+1} \equiv \sum_b \kappa^b)^{\eta-1}(q^b_{t+1})^{-\eta}$. Plugging equation (61) into the budget constraint, equation (49), using equation (60) and defining M_t as

$$M_t \equiv \hat{r}_t k_t + \sum_b (1 + \tilde{r}^b_t) D^b_t - (k_t - (1-\delta)k_{t-1})R_t P_{t-1} + T_t \tag{62}$$

we get

$$Q_{t+1}D_{t+1} + D_{t+1}Q^{\eta}_{t+1}\tilde{Q}_{t+1} = M_t \rightarrow D_{t+1} = \frac{\alpha M_t}{Q_{t+1} +^{\eta}_{t+1} \tilde{Q}_{t+1}}$$
$$\text{and } P_t C^c_t = \frac{Q_{t+1}M_t}{Q_{t+1} + Q^{\eta}_{t+1}\tilde{Q}_{t+1}}.$$

which are equations equation (15) and equation (16) in the main text.

B.2.1 Derivatives at the steady state

Having calculated capitalists' demand for loans and deposits, we calculate the derivatives of these functions with respect to the cost of loans and deposits (r and q respectively). Throughout, we will use the fact that in a steady state, the Euler equation equation (54) becomes

$$1 = \frac{\beta(1-\delta)R_n P_n}{R_n P_n - \hat{r}_n}. \tag{63}$$

Derivative of L with respect to r. The demand function for loans is

$$L^b_{t+1} = P_t i_t(R_{t+1})(\gamma^b)^{\sigma-1}\left(\frac{R_{t+1}}{1+r^b_{t+1}}\right)^{\sigma}.$$

The derivative and elasticity of loans with respect to r are, respectively,

$$\frac{\partial L^b_{t+1}}{\partial r^b_{t+1}} = \underbrace{\left\{\sigma\frac{L^b_{t+1}}{R_{t+1}} + \frac{L^b_{t+1}}{i_t}\frac{\partial i_t}{\partial R_{t+1}}\right\}\left(\frac{R_{t+1}}{1+r^b_{t+1}}\right)^{\sigma}(\gamma^b)^{\sigma-1}}_{\frac{\partial L^b_n}{\partial R_n}\frac{\partial R_n}{\partial r_n}}\underbrace{-\sigma\frac{L^b_n}{1+r^b_n}}_{\frac{\partial L^b_n}{\partial r^b_n}}$$

$$\begin{aligned}
\text{and } \epsilon_L &\equiv -\frac{\partial L^b_{t+1}}{\partial r^b_{t+1}}\frac{1+r^b_{t+1}}{L^b_{t+1}} \\
&= \sigma\left(1-s^b_{t+1}\right) - s^b_{t+1}\times\underbrace{\frac{\partial i_t}{\partial R_{t+1}}\frac{R_{t+1}}{i_t}}_{\equiv-\varepsilon^{i,R}_n} \\
&= \sigma\left(1-s^b_{t+1}\right) + s^b_{t+1}\varepsilon^{i,R}_n, \\
\text{where } s^b_{t+1} &\equiv \left(\frac{R_{t+1}}{1+r^b_{t+1}}\gamma^b\right)^{\sigma-1} = \underbrace{\frac{(1+r^b_{r+1})L^b_{t+1}}{i_t R_{t+1}P_t}}_{\text{Revenue Share}}.
\end{aligned}$$

To calculate the elasticity of investment with respect to the interest rate R, start from the budget constraint equation (49) evaluated at $t+1$ and the Euler equation, equation (54),

$$\frac{P_{t+1}C_{t+1}}{P_t C_t} = \frac{\hat{r}_{t+1}k_{t+1} + \sum_b D^b_{t+1}(1+r^b_{t+1}) + T^c_{t+1} - \sum_b D^b_{t+2} - i_t R_{t+1}P_t}{P_t C_t} = \frac{\beta(1-\delta)R_{t+1}P_t}{R_t P_{t-1} - \hat{r}_t}$$

$$i_t\left(\hat{r}_{t+1} - R_{t+1}P_t\right) + \hat{r}_{t+1}(1-\delta)k_t + \sum_b D^b_{t+1}(1+r^b_{t+1}) + T^c_{t+1} - \sum_b D^b_{t+2} = \frac{\beta(1-\delta)R_{t+1}P_t}{R_t P_{t-1} - \hat{r}_t}P_t C_t$$

$$i_t = \frac{1}{\hat{r}_{t+1} - R_{t+1}P_t}\left(\frac{\beta(1-\delta)R_{t+1}P_t}{R_t P_{t-1} - \hat{r}_t}P_t C_t - \hat{r}_{t+1}(1-\delta)k_t - \sum_b D^b_{t+1}(1+r^b_{t+1}) - T^c_{t+1} + \sum_b D^b_{t+2}\right)$$

$$\frac{\partial i_t}{\partial R_{t+1}} = -\frac{i_t P_t}{R_{t+1}P_t - \hat{r}_{t+1}} - \frac{\beta(1-\delta)P_t}{R_t P_{t-1} - \hat{r}_t}\times\frac{P_t C_t}{R_{t+1}P_t - \hat{r}_{t+1}}$$

Evaluated at the steady state, this expression can be simplified to

$$\begin{aligned}
\frac{\partial i_n}{\partial R_n} &= -\frac{1}{R_n}\times\frac{i_n R_n P_n + P_n C_n}{\beta(1-\delta)R_n P_n}, \\
\rightarrow \epsilon^{i,R}_n &= -\frac{1}{i_n}\times\frac{i_n R_n P_n + P_n C_n}{\beta(1-\delta)R_n P_n} = \frac{1}{\beta(1-\delta)}\left(1 + \frac{D_n Q_n}{\alpha i_n R_n P_n}\right)
\end{aligned}$$

which shows as equation (25) in the main text. Plugging this back into the loan-elasticity,

$$\epsilon_n^{L,r} = \sigma(1 - s_n^b) + s_n^b \epsilon_n^{i,R}.$$

which shows as equation (24) in the main text.

B.3 Banks

We assume oligopolistic competition at the local level in loans, and monopolistic competition on the deposit side. While our framework is amenable to incorporating oilgopolistic competition on deposits, we prefer to assume monopolistic competition given that it is arguably easier for depositors to move their money between banks.

Omitting super-script b to keep notation clean, the problem of a bank is described as

$$\max_{\{\{r_{nt},\tilde{r}_{nt}\},F_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \Big\{ \sum_n L_{nt}(1+r_{nt}) + D_{nt+1} - L_{nt+1} - D_{nt}(1+\tilde{r}_{nt}) + F_{t+1} - \exp\left(\phi \frac{F_t}{\sum_n D_{nt}}\right)(1+r_t^F)F_t \Big\}$$
$$\underline{s.t:}\ [\lambda_t]\ \sum_n L_{nt+1} = \sum_n D_{nt+1} + F_{t+1} \quad \forall t.$$

The first order condition with respect to F_{t+1} reads

$$\beta^t - \beta^{t+1}\left\{\exp\left(\phi\frac{F_{t+1}}{\sum_n D_{nt+1}}\right)(1+r_{t+1}^F) + \exp\left(\phi\frac{F_{t+1}}{\sum_n D_{nt+1}}\right)(1+r_{t+1}^F)\phi\frac{F_{t+1}}{\sum_n D_{nt+1}}\right\} + \lambda_t = 0$$
$$\frac{1}{\beta} + \mu_t = \exp\left(\phi\frac{F_{t+1}}{\sum_n D_{nt+1}}\right)(1+r_{t+1}^F)\left(1+\phi\frac{F_{t+1}}{\sum_n D_{nt+1}}\right) \tag{64}$$

Where $\mu_t = \frac{\lambda_t}{\beta^{t+1}}$. This expression for the marginal cost is equation (21) in the main text.

The first order condition with respect to L_{nt} reads

$$\frac{\partial L_{nt+1}}{\partial r_{nt+1}}\left[\frac{1}{\beta} - (1+r_{nt+1}) + \mu_t\right] = L_{nt+1}$$
$$\epsilon_L\left[-\frac{1}{\beta} + (1+r_{nt+1}) - \mu_t\right] = (1+r_{nt+1})$$

Solving for the interest rate,

$$1 + r_{nt+1} = \frac{\epsilon_L}{\epsilon_L - 1}\left(\frac{1}{\beta} + \mu_t\right) \tag{65}$$

which is equation (22) in the main text. The first order condition with respect to deposits reads

$$\frac{\partial D_{nt+1}}{\partial q_{nt+1}}\underbrace{\frac{\partial q_{nt+1}}{\partial \tilde{r}_{nt+1}}}_{\text{in SS: } -\beta}\left[\frac{1}{\beta} - (1+\tilde{r}_{nt+1}) + \exp\left(\phi\frac{F_{t+1}}{\sum_n D_{nt+1}}\right)(1+r_{t+1}^F)\phi\left(\frac{F_{t+1}}{\sum_n D_{nt+1}}\right)^2 + \mu_t\right] = D_{nt+1}$$
$$\epsilon_D\left[\underbrace{1-\beta(1+\tilde{r}_{nt+1})}_{q_{nt+1}} + \beta\exp\left(\phi\frac{F_{t+1}}{\sum_n D_{nt+1}}\right)(1+r_{t+1}^F)\phi\left(\frac{F_{t+1}}{\sum_n D_{nt+1}}\right)^2 + \beta\mu_t\right] = q_{nt+1}$$

Solving for q

$$q_{nt+1} = -\frac{\epsilon_D}{\epsilon_D - 1}\beta \left\{ \exp\left(\phi \frac{F_{t+1}}{\sum_n D_{nt+1}}\right)(1 + r^F_{t+1})\phi \left(\frac{F_{t+1}}{\sum_n D_{nt+1}}\right)^2 + \mu_t \right\} \tag{66}$$

In equation (23) we have substituted $\epsilon_D = \eta$ given our monopolistic competition assumption.

B.4 Special case in Section 4.3

We fix a city n and omit the subscript n throughout this section. Define

$$y_b(\phi) \equiv 1 + r^b(\phi) \tag{67}$$

$$c_b(\phi) \equiv \mathcal{MC}^b(\phi) \tag{68}$$

$$s_b(\phi) \equiv \frac{y_b(\phi)^{1-\sigma}}{\sum_{v=1}^{B} y_v(\phi)^{1-\sigma}} \tag{69}$$

From equation (22) and equation (24),

$$y_b(\phi) = M(s_b(\phi))c_b(\phi), \text{ where } M(s) \equiv \frac{\sigma - s\Delta}{\sigma - s\Delta - 1} \tag{70}$$

and Δ is defined in equation (4.3)

At the frictionless benchmark $\phi = 0$, and all banks are symmetric within the city, so

$$s_b(0) = \tfrac{1}{B},\ c_b(0) = c_0 \equiv 1 + r^F. \tag{71}$$

Defining also

$$A \equiv \sigma - \tfrac{\delta}{B},\ M_0 \equiv M(\tfrac{1}{B}) = \frac{A}{A-1},\ M_1 \equiv M'(\tfrac{1}{B}) = \frac{\Delta}{(A-1)^2} \tag{72}$$

From here it follows that $y_0 = M_0 c_0$. From equation (21) it follows that

$$\frac{\partial c_b}{\partial \phi}|_{\phi=0} = 2(1 + r^F)\omega_b = 2c_0\omega_b, \tag{73}$$

where, as in the main text, $\omega_b \equiv \frac{F^b}{D^b}$ and $\bar{\omega} \equiv \frac{1}{B}\sum_{v=1}^{B} \omega_v$.

To recover how local average rates depend on ϕ we begin by differentiating bank share with respect to ϕ. Due to variable markups, interest rates depend on market shares which in turn depend on interest rates. To write the system of equations that captures these interactions we begin by looking at market shares.

In logs, shares are

$$\log s_b = (1 - \sigma)\log y_b - \log \sum_{v=1}^{B} y_v^{1-\sigma}. \tag{74}$$

Differentiating,

$$\frac{\partial s_b}{\partial \phi} = (1 - \sigma)s_b\Big[\frac{1}{y_b}\frac{\partial y_b}{\partial \phi} - \sum_{v=1}^{B} \tfrac{s_v}{y_v}\frac{\partial y_v}{\partial \phi}\Big] \tag{75}$$

which evaluated at the symmetric benchmark yields

$$\frac{\partial s_b}{\partial \phi}|_{\phi=0} = \frac{(1-\sigma)}{By_0}\Big[\frac{\partial y_b}{\partial \phi}|_{\phi=0} - \frac{\partial \bar{y}}{\partial \phi}|_{\phi=0}\Big] \;\; \bar{y} \equiv \tfrac{1}{B}\sum_{v=1}^{B} y_v \tag{76}$$

Differentiating equation (70),

$$\frac{\partial y_b}{\partial \phi} = M(s_b)\frac{\partial c_b}{\partial \phi} + c_b M'(s_b)\frac{\partial s_b}{\partial \phi}. \tag{77}$$

At $\phi = 0$,

$$\frac{\partial y_b}{\partial \phi}|_{\phi=0} = M_0\frac{\partial c_0}{\partial \phi}|_{\phi=0} + c_0 M_1\frac{\partial s_b}{\partial \phi}|_{\phi=0}. \tag{78}$$

Using equation (76),

$$\frac{\partial y_b}{\partial \phi}|_{\phi=0} = M_0\frac{\partial c_0}{\partial \phi}|_{\phi=0} + \frac{(1-\sigma)c_0 M_1}{By_0}\Big[\frac{\partial y_b}{\partial \phi}|_{\phi=0} - \frac{\partial \bar{y}}{\partial \phi}|_{\phi=0}\Big]. \tag{79}$$

Defining

$$\lambda \equiv \frac{(1-\sigma)c_0 M_1}{By_0} = \frac{(1-\sigma)M_1}{BM_0} = \frac{(1-\sigma)\Delta}{BA(A-1)}, \tag{80}$$

equation (79) becomes

$$(1-\lambda)\frac{\partial y_b}{\partial \phi}|_{\phi=0} + \lambda\frac{\partial \bar{y}}{\partial \phi} = M_0\frac{\partial c_0}{\partial \phi}|_{\phi=0}. \tag{81}$$

Taking the average of this expression across banks,

$$\frac{\partial \bar{y}}{\partial \phi}|_{\phi=0} = M_0\frac{\partial \bar{c}}{\partial \phi}|_{\phi=0} \tag{82}$$

where $\bar{c} \equiv \frac{1}{B}\sum_{v=1}^{B} c_v$. Using equation (73),

$$\frac{\partial \bar{c}}{\partial \phi}|_{\phi=0} = \tfrac{1}{B}\sum_{v=1}^{B} 2c_0\omega_v = 2c_0\bar{\omega} \tag{83}$$

and, therefore,

$$\frac{\partial \bar{y}}{\partial \phi}|_{\phi=0} = 2M_0 c_0\bar{\omega} \tag{84}$$

Substituting back into equation (81),

$$\frac{\partial y_b}{\partial \phi}|_{\phi=0} = \frac{M_0}{1-\lambda}(\omega_b - \lambda\bar{\omega}) = 2y_0\Big(\bar{\omega} + \frac{\omega_b - \bar{\omega}}{1-\lambda}\Big). \tag{85}$$

From here it follows that

$$\frac{\partial y_b}{\partial \phi}|_{\phi=0} = 2y_0\Big(\bar{\omega} + \tfrac{\omega_b - \bar{\omega}}{\Xi}\Big) \tag{86}$$

and

$$\frac{\partial s_b}{\partial \phi}|_{\phi=0} = \frac{2(1-\sigma)}{B}\frac{\omega_b - \bar{\omega}}{\Xi}, \tag{87}$$

where the dampening factor Ξ is defined as

$$\Xi \equiv 1 - \lambda = 1 + \frac{(\sigma-1)\Delta/B}{A(A-1)} = 1 + \frac{(\sigma-1)\Delta/B}{(\sigma-\Delta/B)(\sigma-\Delta/B-1)}. \tag{88}$$

Finally, we can calculate changes in the average interest rate,

$$y_n^w(\phi) \equiv \sum_{b=1}^{B} y_b(\phi) s_b(\phi) \tag{89}$$

$$\frac{\partial y_n^w(\phi)}{\partial \phi}|_{\phi=0} = y_0 \overbrace{\sum_{b=1}^{B} \frac{\partial s_b}{\partial \phi}}^{=0} + \frac{1}{B}\sum_{b=1}^{B} \frac{\partial y_b}{\partial \phi}|_{\phi=0} = 2y_o\bar{\omega} \tag{90}$$

which leads to the expression in the main text.

C Estimation Appendix

C.1 Estimation algorithm

Our estimation strategy proceeds in two stages. In the outer loop, we search over structural parameters $\Gamma^o \equiv \{\sigma, \phi\}$. For each candidate value of $\{\sigma, \phi\}$, the inner loop inverts the model to recover $\Gamma^I \equiv \{z_n, b_n, \{\gamma_n^b, \kappa_n^b\}_{b\in\mathcal{C}^b}\}_{n=1}^N$. Our estimated Γ^o is chosen to match the estimated effects of deposit shocks on loan quantities and interest rates from Section 3.

Inner loop For a given pair $\{\sigma, \phi\}$, we recover the model's implied fundamentals Γ^I in four steps.

Step 1: City-bank match parameters. We estimate the city-bank match parameters $\{\gamma_n^b\}$ and $\{\kappa_n^b\}$ to exactly replicate observed loan and deposit volumes at the city-bank level in 2015.[22]

In the data we observe total loan repayment in each city,

$$\mathcal{L}(i_n) = \sum_{b\in\mathcal{B}^n} (1+r^b) L_n^b.$$

In the model, loan repayment equals investment expenditure: $\mathcal{L}(i_n) = i_n R_n P_n$. Using the loan demand function, equation (**??**), we can write loan volumes as a function of the match parameters

$$L_n^b = \mathcal{L}(i_n) \frac{R_n^{\sigma_0-1}}{(1+r_n^b)^{\sigma_0}} (\gamma_n^b)^{\sigma_0-1}.$$

This yields a system of $\tilde{N}$ equations in $\tilde{N}$ unknowns (where $\tilde{N}$ is the number of city-bank pairs operating in Chile in 2015). Solving this system delivers estimates $\{\hat{\gamma}_n^b\}$ that perfectly rationalize observed city-bank loan volumes.

To estimate κ_n^b, we use data on average deposit rates by bank in 2015. Under our assumption of monopolistic competition in the deposit market, banks offer uniform deposit rates across all cities they serve. In steady state, equation equation (56) implies that the ratio of deposits from two banks in the same city satisfies:

$$\frac{D_n^b}{D_n^{b'}} = \left(\frac{\kappa^b}{\kappa^{b'}}\right)^{\eta-1} \left(\frac{\beta-(1+r_n^b)}{\beta-(1+r_n^{b'})}\right)^{-\eta}. \tag{91}$$

For each city n, we normalize $\kappa_n^b = 1$ for one bank, then use equation equation (91) to solve for κ_n^b for all other banks operating in that city. Finally, we normalize the city-level average of κ_n^b to equal one.

[22] We cannot extract information specific to a bank from the micro-data, so in this section we use publicly available data on new loans by city-bank, the average interest rate by bank, and the average interest rate by city from the CMF.

Step 2: Free-on-board prices. Given the estimated match parameters, we solve for the vector of (unobserved) free-on-board prices $\{p_n\}$ that rationalizes the observed spatial distribution of wages and employment as an equilibrium. This requires imposing goods market clearing in all N cities. We normalize the average price of local goods to one, yielding N independent equations in N unknowns.

Step 3: Local productivities. With prices $\{\hat{p}_n\}$ in hand, we back out city-specific productivities from the zero-profit condition:

$$\hat{z}_n = \frac{w_n^{\mu} \hat{r}^{1-\mu}}{\hat{p}_n}, \tag{92}$$

where w_n is observed in the data and $\hat{r}_n$ is the estimated marginal product of capital in city n. We recover $\hat{r}_n$ from equation (63) using our estimates of γ_n^b from Step 1, which allow us to compute the loan price indices R_n and P_n.

Step 4: Local amenities. Finally, we recover amenities $\{\hat{b}_n\}$ as the values that rationalize the observed distribution of workers across cities as a migration equilibrium. These are pinned down by the labor market clearing condition, equation (29), where all other terms are now known. We normalize the average amenity to equal one.

Outer loop We search over the structural parameters $\Gamma^o = \{\sigma, \phi\}$ to match our reduced-form empirical estimates from Section 3.

To construct model-based analogs of our empirical moments, we proceed as follows. We increase productivity by 1% in the fishing cities (those with high employment shares in fishing). For each candidate $\{\sigma, \phi\}$, we solve for the new equilibrium and implement the empirical strategy used in Section 3.

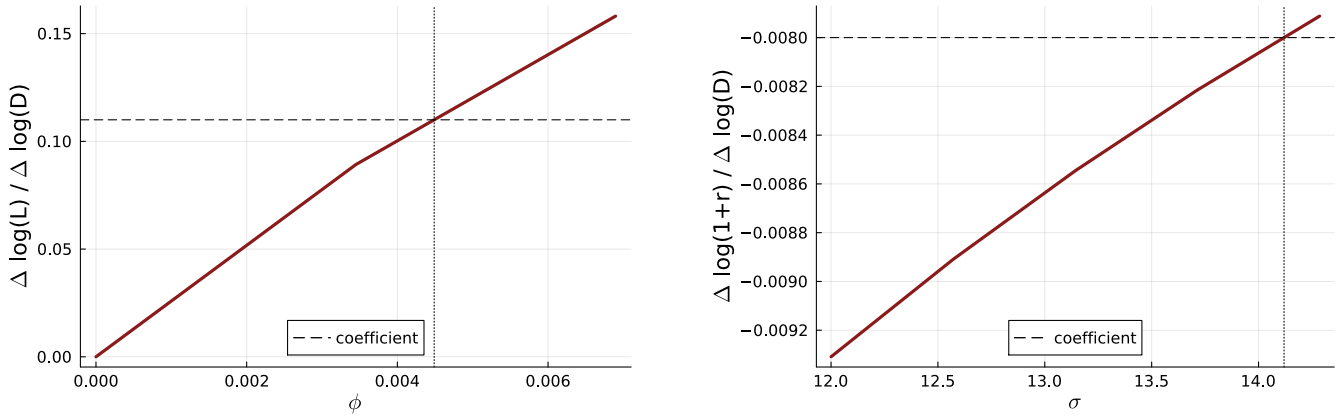
Specifically, we construct bank-level exposure to the shock as in equation (2) as the share of each bank's baseline deposits originating in fishing cities. We then run a first-stage regression at the bank level of the change in log total deposits on this exposure measure. Using predicted deposit growth from this first stage, we estimate second-stage regressions at the city-bank level:

- *Quantity response:* Change in log loans at the city-bank level.
- *Price response:* Change in $\log(1 + r_n^b)$ at the city-bank level.

These second-stage specifications correspond to equation (1) and equation (3) in the main text. By taking differences at the city-bank level before and after the shock, we effectively control for the analog of city fixed effects.

Identification of the structural parameters in Γ^o can be interpreted as follows. The interbank friction parameter, ϕ, primarily governs the magnitude of the quantity response: larger frictions force banks to rely more heavily on retail deposits, amplifying the effect of deposit shocks on lending. The elasticity of substitution, σ, primarily determines the price response: higher elasticity means banks can expand lending with smaller interest rate reductions.

We are able to match the two empirical moments exactly. Figure 16 shows the relationship between each parameter and the targeted empirical elasticities, with dashed lines indicating our preferred estimates. As discussed in the paper, the left panel illustrates that the elasticity of loans with respect to deposits is zero when there are no interbank frictions ($\phi = 0$): in that case, deposits flow freely through the interbank market to the best lending opportunities, so the identity of the shocked bank is irrelevant.



(a) Estimating ϕ

(b) Estimating σ

Figure 16: Estimation strategy, inner and outer loop

C.2 Estimated city-bank match γ_n^b

Figure 17 shows the estimated value of city-bank matches.

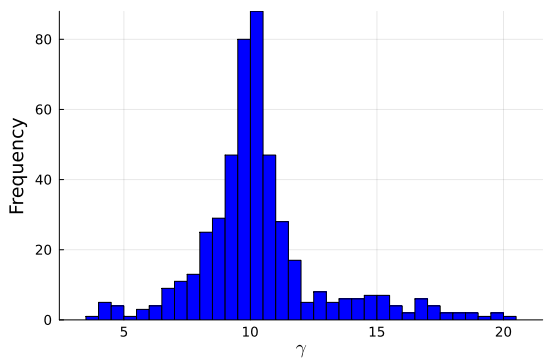


Figure 17: Estimated city-bank match γ

These estimates are closely related to the number of branches that bank b has in city n, after controlling for city and bank fixed effects. Table 10 below shows the results of an OLS estimate of

$$Log(\gamma_n^b) = \beta_0 + \gamma_n + \gamma_b + \beta_1 Log(Branches_n^b) + \varepsilon_n^b. \tag{93}$$

Table 10: Branches

	Estimated city-bank match (Log)
Branches	0.03***
	(0.006)
Bank FE	Yes
City FE	Yes
Observations	344